Same Crime, Different Time:

Evidence from DWI Cases that Extralegal Factors Matter[*]

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Abstract

We document disparities in criminal case outcomes that are driven by extralegal factors as opposed to differences in underlying criminal conduct. We focus on Driving While Intoxicated (DWI) cases, where breath alcohol content (BrAC) is a central piece of evidence. Using administrative data from Harris County, Texas, we condition on BrAC and restrict the sample to first-time offenders above the legal limit without aggravating circumstances, thus holding fixed the legally-relevant facts of the case. Nevertheless, we find large gaps in case dismissal and incarceration along four dimensions: race/ethnicity, gender, attorney type, and neighborhood income. Each dimension matters conditional on the others, and the gaps accumulate. These gaps are too large to be plausibly explained by unobserved differences in driving behavior. They likely reflect both pre-existing disadvantage that shapes defendants' choices post-arrest and differential treatment by judges and prosecutors. Gaps are largest in the most lenient courtrooms, where the benefits of leniency accrue disproportionately to advantaged defendants.

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1 Introduction

Social and economic inequalities are pervasive in the criminal legal system. Disparities by race, ethnicity, gender, and economic status exist at many stages in the legal process and across many jurisdictions. One explanation for these widespread group differences is that they are exclusively driven by differences in criminal conduct. Alternatively, disparities may arise among people with the same criminal conduct because they differ in extralegal factors (i.e., characteristics that are not legally relevant for guilt). The fundamental challenge to distinguishing between these explanations is that objective measures of criminal conduct are rarely observed.

We address this challenge by studying Driving While Intoxicated (DWI) offenses. In this setting, breath alcohol content (BrAC) is a precise, objective measure of the defendant's guilt and the severity of their criminal conduct. By law, a person is guilty of DWI if they drive with a BrAC above a set threshold.[1] This legal standard provides both an empirical opportunity and a normative benchmark: among first-time offenders with no aggravating circumstances and the same BrAC, the law treats all defendants as equally guilty. If case outcomes were determined solely by legally-relevant factors, demographic and socioeconomic characteristics would be unrelated to punishment after conditioning on BrAC. Departures from this benchmark — regardless of their source — constitute evidence that extralegal factors shape who bears the costs of the criminal justice system. In addition to this normative utility, DWIs are of interest because they are one of the most common ways that people interact with the justice system, representing 10% of all arrests in the U.S. (Federal Bureau of Investigation, 2019).

Drawing on administrative data from Harris County, Texas, the third most populous

[1] We use DWI as legal shorthand for the arrests and convictions drivers receive for driving while impaired or under the influence of alcohol. Depending on the state, the legal statute may refer to driving while intoxicated (DWI), driving under the influence (DUI), operating under the influence (OUI), or operating a vehicle while impaired (OVI). These legal statutes can largely be regarded as equivalent, and the acronym used is usually defined locally based on the relevant legal statute.

county in the U.S. and home to Houston, we show that extralegal factors affect criminal case outcomes. We begin with simple between-group comparisons. Restricting attention to first-time offenders with no prior DWI arrests, no aggravating circumstances, and the same BrAC, we compare outcomes along four dimensions: race/ethnicity, gender, use of a private attorney, and neighborhood median income.[2] Despite sharing the same legally-relevant case characteristics, people from different demographic and socioeconomic backgrounds face substantially different outcomes. Among defendants just above the legal limit, white defendants are nearly fifty percent more likely to have their case dismissed compared to Hispanic defendants (60% vs. 40%). Defendants from neighborhoods in the top quartile of the income distribution are over sixty percent more likely to have their case dismissed than those from the bottom quartile (77% vs. 47%).

The dimensions we consider are not independent, and it may be the case that disparities by race simply reflect differences in access to economic resources across these groups. However, despite the highly correlated nature of these factors, we find that the disparities across groups aggregate to larger cumulative disparities. To illustrate this, we first provide a striking example. Looking at cases just above the legal limit, a white, female defendant with a private attorney and from a neighborhood in the top half of the income distribution is nearly twice as likely to have her case dismissed as a Hispanic male without a private attorney and from a neighborhood in the bottom half of the income distribution. This relative advantage persists even for defendants whose BrAC is two or three times the legal limit.

We formalize the idea of cumulative disparities by estimating a joint regression including all dimensions at once. We find that each factor contributes to the case outcome, even conditional on all other factors. Put differently, a Hispanic defendant is less likely to have their case dismissed than a white defendant, even when both are male, have the

[2] We treat attorney type as an extralegal factor because the law treats defendants equally regardless of representation. However, the choice to hire a private attorney is also a means through which other extralegal factors may translate into outcomes.

same BrAC, have private attorneys, and are from neighborhoods with similar median income. Similarly, defendants with different economic resources face different outcomes even when they share the same demographic characteristics and have retained private lawyers.

We argue that the documented disparities are unlikely to be the consequence of unobserved differences in driving behavior that are legally relevant to case outcomes. While we do not observe the speed or recklessness of driving leading up to each breath test, the gaps in outcomes across groups are too large to be plausibly explained by within-BrAC differences in driving. For example, to rationalize the observed dismissal gaps, defendants from the most disadvantaged group with a BrAC of 0.08 would need to be driving as dangerously as those from the most advantaged group with a BrAC of 0.15—a difference in intoxication large enough to more than quadruple the risk of a fatal crash. At slightly higher BrAC levels for the disadvantaged group, no level of intoxication among the advantaged group yields comparable punishment. These patterns mean that differences in driving behavior cannot plausibly explain the disparities we document. We discuss these and other threats to comparability in detail in Section 7.

In the last section of the paper, we consider the channels through which extralegal factors could influence judicial outcomes. Holding criminal conduct fixed, the gaps in judicial outcomes must reflect either direct discrimination by system actors, differences in defendant decision-making post-arrest, or both. Our paper's core contribution is to show that disparities in case outcomes are not due to differences in offense severity alone, and thus, that one or both of these forces are at work. While we cannot speak to the relative importance of each channel, we present evidence that is consistent with both contributing to the observed disparities.

First, we consider that, even when presented with the same menu of legal options, defendants from different socioeconomic backgrounds may make different choices. To see why, it helps to understand the three tracks that DWI cases in our sample follow.

These tracks—dismissal, probation, and incarceration—differ in their monetary, time, and stigma costs. Those who end up with a dismissal either have their cases dismissed outright or after satisfying the conditions of a pretrial diversion program. Defendants whose cases are dismissed avoid a criminal conviction, which can adversely affect employment, housing, and other opportunities. Those sentenced to probation are similarly expected to satisfy court-mandated conditions to avoid jail time, such as using an ignition interlock, cooperating with community supervision, completing driving school, and paying fees and fines. Defendants sentenced to incarceration are less likely to receive financial sanctions and ultimately serve less time than probationers who fail to meet the required conditions.

Although probation and pretrial diversion are often framed as more lenient or reform-oriented, they are associated with higher financial burdens, including fines, supervision fees, and treatment costs. They also carry the risk of a relatively longer jail sentence. Moreover, since time spent in pre-trial detention is typically credited against a jail sentence, a defendant's ability to make bail can materially alter the relative appeal of incarceration versus alternatives. For a person with limited financial means or job flexibility, a short jail sentence may be preferable to a prolonged and costly period of probation. In this way, differences in defendant decisions likely explain some of the disparities we observe.

Beyond defendant decision-making, system actors themselves appear to influence the disparities. Conditional on defendant characteristics, gaps in dismissal and incarceration rates vary substantially across courtrooms[3]—indicating that which courtroom a defendant is assigned to meaningfully shapes their outcome, independent of their own choices or circumstances. Different offers to different groups could be justified on utilitarian grounds if courtrooms engage in accurate statistical discrimination—that is, if decision-makers correctly identify higher-risk defendants and assign sanctions that actually reduce future offending. We find no evidence that this is the case; if anything, we present sugges-

[3] In this setting, courtroom assignment determines which judge and prosecutor will handle the case.

tive evidence from instrumental variables specifications using judge harshness that more punitive sanctions appear to increase recidivism rather than reduce it.

We find that disparities in incarceration and conviction are smallest where courtrooms are harshest—that is, where dismissal rates are lowest and incarceration rates highest. This pattern suggests that although lenient courts may offer more favorable outcomes, those benefits disproportionately accrue to defendants with the resources to capitalize on them. Alternatively, lenient courtrooms may be more likely to reserve favorable dispositions for advantaged defendants. In either case, the findings warn that judicial or prosecutorial discretion may not reliably serve as a mechanism for leveling the playing field across defendants. To the extent that gaps instead reflect advantaged defendants being better positioned to capitalize on the full range of options available in lenient courtrooms, this underscores the importance of the systemic disadvantage outlined above.

The remainder of the paper proceeds as follows. Section 2 discusses our work in relation to existing work. Section 3 describes relevant institutional details about DWIs. Section 4 describes our data. Section 5 discusses the types of punishments that are experienced after a DWI arrest. Section 6 documents the disparities by defendant characteristics, even after controlling for objective measures of severity. Section 7 discusses what might undermine interpretation of these disparities as evidence of the influence of extralegal factors. Section 8 discusses the ways in which extralegal factors could affect case outcomes, including defendant decision-making and system actor behavior, and provides suggestive evidence that the disparities are not generated by accurate statistical discrimination. Lastly, Section 9 concludes.

2 Related Literature

A large and growing body of research documents disparities at various stages of the criminal justice process. Much of this work uses quasi-experimental variation or additional structure to isolate specific disparity-generating mechanisms, such as the role of decision-maker identity (Antonovics and Knight, 2009; Anwar et al., 2012; Eren and Mocan, 2018; Sloan, 2024) or the effect of specific institutional features or policies on observed gaps (Grogger and Ridgeway, 2006; Horrace and Rohlin, 2016; Goncalves and Mello, 2021; Tuttle, 2025; Fischman and Schanzenbach, 2012; Rose, 2021), or to draw inferences about the nature or justifiability of disparities (Arnold et al., 2018, 2022; Feigenberg and Miller, 2022, 2025). These designs achieve identification through variation in the system, such as who the decision-maker is or where a policy threshold falls, while typically taking the composition of the subject pool — and any pre-existing differences across groups — as given. As a result, they are well-suited to establishing that a particular disparity-generating mechanism is at work, but less suited to characterizing the total disparity in outcomes experienced by subjects who enter the system with comparable case facts but different demographic or socioeconomic characteristics. Our paper takes a complementary approach. By conditioning on an objective, precisely measured indicator of offense severity among otherwise comparable defendants, we measure the cumulative size of the disparities created by extralegal factors along multiple dimensions: race/ethnicity, gender, attorney type, and socioeconomic status.

The papers most closely related to ours also estimate disparities after conditioning on observable measures of offense severity. In studies of federal sentencing, Mustard (2001), Rehavi and Starr (2014), and Starr (2015) show that gaps across demographic groups persist even after controlling for criminal history, charge offense level, and arrest offense. Stevenson (2018) takes a similar approach to pretrial detention, showing that roughly a third of the racial gap can be attributed to differences in ability to pay bail. These findings suffer from the limitation that the variables conditioned on — charge, arrest offense, criminal history, bail amount — do not capture all the key information about the underlying facts of the case and are subject to the discretion of system actors. Aggarwal et al. (2025) overcome this limitation in the context of traffic stops by using high-frequency location data to infer driver speed, showing that non-white Lyft drivers are more likely to be stopped than white drivers even when engaging in comparable driving behavior. Our setting offers a similar advantage: BrAC provides an objective measure of offense severity that is strongly predictive of case outcomes. Moreover, the BrAC readings in our data are the same evidence observed by the police, prosecutors, and judges handling these cases, rather than a researcher-constructed proxy for the information available to decision-makers. Unlike studies of traffic stops, we also observe the full adjudication process, allowing us to document disparities in criminal case outcomes — dismissal, probation, and incarceration — across multiple defendant characteristics simultaneously.

Kunitz et al. (2006) also study DWI outcomes in a New Mexico county, conditioning on a linear term in blood alcohol content. We extend this work with a larger and more recent sample, a broader set of defendant characteristics including attorney type and economic resources, and a tighter comparison group restricted to first-time offenders above the legal threshold with no aggravating circumstances.[4]

3 Institutional Setting

DWI is one of the most prevalent and deadly offenses in the United States. According to the Centers for Disease Control and Prevention (2023), approximately 18.5 million people drove while impaired by alcohol during 2020, representing 7.2 percent of people 16 years or older. This led to 11,654 deaths in alcohol-related crashes, which is roughly half as many as the number of homicides during the same period. When considering broader externalities, this expands to over 338,000 injuries in 2020. In addition to the human cost of this behavior, it also led to high levels of interaction between drivers and the criminal justice system. According to the Federal Bureau of Investigation (2019), over 1 million people were arrested under suspicion of DWI in 2018. This is nearly twice as many as all violent crimes and about 60 percent as large as all drug-related offenses. As we see in our

[4] As in most states, prior DWI convictions influence the ranges of jail time, fines and license suspensions the DWI convicted face.

data, DWI arrests span all demographic groups, affecting people across a wide range of racial, economic, and social backgrounds.

The criminal justice process for people suspected of DWI begins with police observation. An officer may initiate a traffic stop either due to a separate traffic violation, such as speeding or running a red light, or because of specific indicators of impairment, like swerving, erratic driving, or delayed reaction times.[5] During the stop, additional cues such as the odor of alcohol or an open container in the vehicle may further raise suspicion. The officer may administer field sobriety tests to assess physical and cognitive impairment. Upon establishing probable cause, the officer can request a chemical test, typically a breathalyzer or blood draw, to measure the driver's alcohol concentration. Under Texas Penal Code § 49.04 (Texas Legislature, 2023b), it is a criminal offense to operate a motor vehicle in a public place while intoxicated, defined in § 49.01(2) as lacking the normal use of mental or physical faculties due to alcohol or drugs, or having a BrAC of 0.08 or higher. Refusing a chemical test results in automatic administrative license suspension and may be used as evidence at trial. In such cases, officers frequently obtain a warrant to collect a blood sample.[6]

Once chemical test results are available, the case is referred to the district attorney, who determines whether to charge the driver with DWI. In addition to BrAC, the law specifies several other factors as relevant to the severity of a DWI offense. These include history of prior DWIs, having an open container of alcohol in the vehicle, the presence of a child passenger, and causing injury or death. Once charged, defendants appear before a judge for a bail hearing. Those who post bail or are released on their own recognizance await further proceedings out of custody, while others remain incarcerated. All defendants are entitled to legal representation.

[5] In Texas, DWI checkpoints are banned, and so deterrence through these salient and unpredictable stops is not allowed (Banerjee et al., 2021; Matsuzawa, 2025).

[6] According to Jones and Nichols (2012), approximately 20 percent of defendants refuse to undergo a breath test. We do not observe such individuals in our dataset.

4 Data

Our study focuses on defendants who underwent a breath alcohol test in Harris County, Texas, in 2004 and from 2009 through the first half of 2015.[7] These records were obtained via a public records request to the state of Texas. We restrict attention to people with BrAC measures at or above 0.08, the statutory cutoff for DWI, who appear in Harris County court records. Figure A.1 shows the probability of a formal DWI charge as a function of BrAC. Some drivers below the 0.08 threshold are still charged, likely due to behavioral evidence of impairment or the presence of non-alcohol substances. Above the 0.08 threshold for *per se* intoxication about 90 percent of drivers are charged. Since charging rates appear unrelated to BrAC beyond this point, the remaining cases are likely missing at random, possibly due to data merge errors.

To ensure that we are making comparisons between groups that share all legally-relevant factors, we further limit our sample in two important ways. First, we restrict our sample to drivers with a first-time DWI offense, with no prior criminal convictions, and with no aggravating circumstances (such as a minor being present in the car).[8] Second, we remove people who are not U.S. citizens per the court records.[9] These restrictions yield an analysis sample of 12,888 defendants. For a subset of 759 defendants, we collected additional detailed information on case outcomes directly from the case dockets on the Harris County Courts website. Additional details about the sample and data construction are available in Appendix B.

[7] We also requested breath tests for the years 2005-2008, years earlier than 2004, and years after 2015. Ultimately, only records for 2004 and 2009-2020 were available. The records from 2016-2020, however, do not contain the driver's first name. This window includes a period in which driver responsibility fees were held constant in Texas (Finlay et al., 2023).

[8] We can reliably observe prior DWIs and all other convictions in Harris County for 14 years. However, if a defendant has prior offenses outside of Harris County, we do not observe them. In supplemental analyses, we further remove defendants who match to the statewide convictions database, on the basis of name and age, at any point prior to the focal DWI case.

[9] We do not consider this group in our main analyses because of the distinct incentives faced by undocumented immigrants and the ambiguity surrounding their legal protections in criminal proceedings. We report results comparing outcomes for citizens and non-citizens in Appendix Figure A.6.

Table A.1 presents summary statistics for our sample, highlighting substantial racial and socioeconomic diversity among DWI defendants. Defendant characteristics also vary systematically across sentencing outcomes. Incarceration, for example, falls disproportionately on disadvantaged groups, including Hispanic men. Legal representation is another important dimension of inequality. Counsel plays a central role in plea negotiations, and although defendants retain the right to trial, only 1.5 percent of cases proceed that far; nearly all are resolved through plea bargaining. In our sample, about one fifth of defendants rely on court-appointed attorneys while the remainder are represented by private counsel.

5 Criminal Justice Consequences

Among people charged with DWI, the majority follow one of three mutually exclusive tracks: incarceration, probation, or dismissal. Each path differs in terms of severity, duration, financial burden, and long-term implications. While immediate incarceration primarily serves a punitive and deterrent function, probation and pre-trial diversion incorporate explicit rehabilitative goals. These latter alternatives aim to reduce recidivism by addressing underlying issues such as substance abuse, while still maintaining accountability through structured judicial supervision. Table 1 describes the frequency and severity of judicial outcomes by disposition type in our Harris County analysis sample.

One common outcome is immediate incarceration, typically involving a short jail sentence. In our Harris County sample, a third of first time DWI offenders are incarcerated with an average sentence of about two weeks. In practice, actual time served may be less than the formal sentence due to overcrowding in jail facilities. To minimize disruption to employment, sentences may be structured to allow weekend-only incarceration. Incarcerated individuals may also be required to pay court fines. About half of defendants sentenced to incarceration pay a fine, and the average fine (including zeros) is $380. These cases are resolved quickly on average with the time to final disposition averaging

Table 1: Case Outcomes by Disposition Type

	Dismissed	Probation	Incarcerated
Panel A. Full Sample			
Time to Disposition (Months)	12.779	4.441	3.754
	(4.918)	(3.258)	(3.819)
Probation Duration (Months)	0	12.503	0.024
	-	(2.408)	(0.483)
Any Fine	0	0.934	0.487
	-	(0.249)	(0.500)
Fine Amount	0	526.570	380.185
	-	(655.687)	(468.751)
Suspended Sentence (Months)	0	7.204	0
	-	(2.673)	-
Immediate Sentence (Months)	0	0	0.588
	-	-	(0.740)
Observations	2,811	5,836	4,241
Panel B. Detailed Sample			
License Suspension	0	0.048	1
	-	(0.214)	-
Driving School	0	0.969	0.003
	-	(0.173)	(0.055)
Ignition Interlock	0.308	0.641	0.274
	(0.463)	(0.480)	(0.447)
Observations	144	345	270

Notes: Entries display means and standard deviations for case outcomes by disposition type in our Harris County analysis sample. Panel A reports statistics for outcomes observed in our full sample. Panel B reports statistics for the subset of outcomes observed in the sample for which we obtained detailed information directly from individual case dockets.

only 3.75 months from the time of the incident.

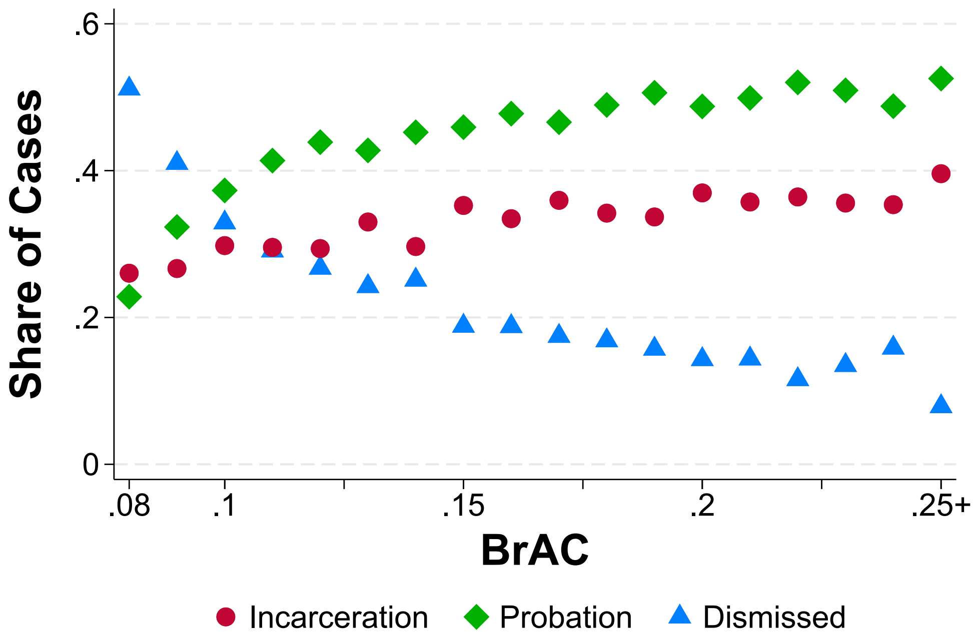
A second group of defendants is sentenced to supervised probation, which imposes ongoing obligations for up to 24 months (Texas Legislature, 2023a). Forty-five percent of the defendants in our sample are sentenced to probation. In addition to fines that average $527 in our sample, probation requires monthly community supervision fees (which range from $25 to $60 in Texas, Texas Legislature (2023a)). Defendants must demonstrate progress toward a court-mandated rehabilitation plan, which may include participation in programs such as driving school and Alcoholics Anonymous. In our sample, nearly all defendants who receive probation are required to attend driving school. Most also install an ignition interlock device on their car. According to Harber Law (2025), these normally cost \$70 to \$150 to install and between \$60 to \$80 each month to rent. Probation sentences typically include a suspended jail term that will not be imposed if the defendant successfully completes the probation period. However, if a defendant fails to meet the terms of supervision, probation may be revoked, and the suspended jail sentence enforced. The suspended jail sentence averages over seven months in our sample.

Twenty-two percent of cases in the sample are ultimately dismissed — either outright or following successful completion of a pretrial diversion program. Diversion closely resembles probation in terms of conditions and oversight, but carries a key advantage: successful completion does not result in a criminal conviction. Like probation, it typically involves supervision fees (though these can be waived for low-income defendants) and regular court interaction for over a year on average before the case is resolved. However, unlike probation or incarceration, dismissal leaves no record of criminal conviction, preserving access to employment, professional licensing, housing, and educational opportunities that are routinely foreclosed by a conviction.

We do not directly observe diversion assignment in the full sample, but in the detailed sample, 61.8 percent of dismissals are resolved through diversion. Since time to disposition is highly predictive of dismissal type, we proxy for outright dismissal in the full sample using an indicator for resolution within one year of the breath test; dismissals taking longer are assumed to follow diversion completion, while those that take less than a year are assumed to be outright dismissals. Among the detailed sample, this strategy correctly categorizes 92.7% of cases resolved in under a year and 86.4% of those resolved after more than a year.

Figure 1 shows the probability of dismissal, probation, and incarceration at each level of BrAC in our sample. As the severity of the offense, measured by BrAC, increases, the probability of dismissal falls from over 50 percent for individuals just above the legal limit to less than 10 percent for individuals with BrAC in excess of 0.25, while the probability of incarceration rises from about 26 percent to 40 percent over the same range. This relationship suggests that prosecutors view incarceration as a harsher penalty, fit for a more serious offense than either probation or pre-trial diversion. Conversely, pre-trial diversion and outright dismissal are disproportionately reserved for comparatively minor offenses.

Figure 1: Case Outcomes by BrAC



Notes: Figure displays the share of cases in each disposition category at every level of BrAC.

6 Disparities in Criminal Justice Outcomes

While it seems both fair and intuitive that more serious offenses should result in more severe sanctions, we show that defendant characteristics unrelated to the offense are also strongly predictive of outcomes. This is where our setting and data allow us to make comparisons among defendants who are similar in factors relevant for guilt. In particular, we will show that defendant characteristics are strongly predictive of outcomes after controlling flexibly for a precise measure of intoxication among first time offenders with no aggravating circumstances. We begin by comparing dismissal and incarceration rates across groups defined by a single characteristic, and then show—both visually and with multivariate regression—that the disadvantages associated with race, gender, income, and legal representation layer on one another.

Panel A of Figure 2 plots dismissal probabilities across demographic and socioeconomic groups at different BrAC levels.[10] The top left subfigure displays dismissal rates by race. At nearly every BrAC level, Black and Hispanic defendants (pooled) have systematically lower dismissal probabilities than white defendants. Just above the 0.08 threshold, white defendants are about 50 percent more likely than Black and Hispanic defendants to have their cases dismissed.

Disparities extend beyond race. Defendants with private attorneys just above the threshold are three times as likely to have their cases dismissed as those with court-appointed counsel, and the gap is statistically significant across the full BrAC range. A defendant with a private attorney and a BrAC twice the legal limit has a similar likelihood of dismissal as a defendant with a court-appointed attorney just above the threshold. Dismissal rates are also significantly higher for defendants from census tracts in the top within-sample income quartile than for those from the bottom quartile. Female defendants generally have higher dismissal rates than males.

In Appendix Figure A.4, we show the rates at which each group receives outright dismissal and dismissal after diversion. The patterns in outright dismissal are qualitatively similar to those we see when looking at dismissal overall. Just above the 0.08 threshold, white defendants are about twice as likely to receive outright dismissal as Black and Hispanic defendants, while a defendant with a private attorney is about three times as likely to receive an outright dismissal as one with a court-provided attorney. Outright dismissal rates for men and women are quite similar. Gaps in dismissal after diversion are less pro-

[10]While the main-text results focus on citizens, Appendix Figure A.6 and Table A.4 show that disparities are even wider when non-citizens are included.

nounced across race and neighborhood income groups, and clearer for groups defined by attorney type and sex.

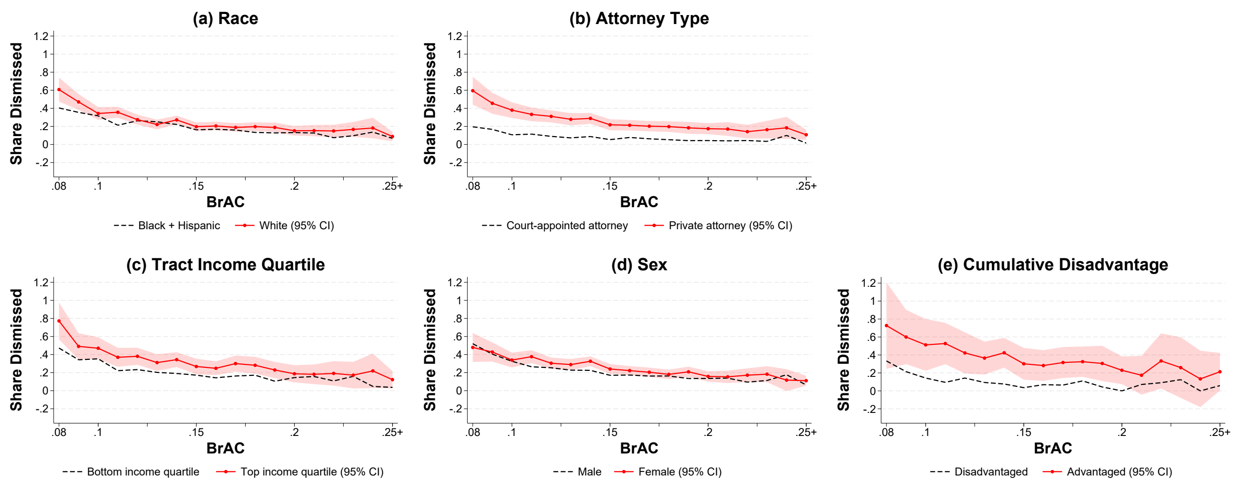
Panel B of Figure 2 shows that similar disparities appear in incarceration. Black and Hispanic defendants (pooled) are about 8 percentage points more likely to be incarcerated than white defendants with the same BrAC. The gap between individuals represented by court-appointed attorneys and those represented by private attorneys is even larger, at about 30 percentage points. Incarceration rates are substantially higher among defendants from the bottom within-sample income quartile than among those from the top quartile. Men are substantially more likely to be incarcerated than women.

Of course, the characteristics we consider are likely correlated with each other. For example, race may be correlated with access to a private attorney, which represents one channel through which a defendant might experience disadvantage. Even if the observed disparities across demographic groups operate entirely through correlated factors—such as Black and Hispanic defendants being less likely to have private counsel—they still reveal that some groups systematically lack access to the most favorable criminal justice outcomes.

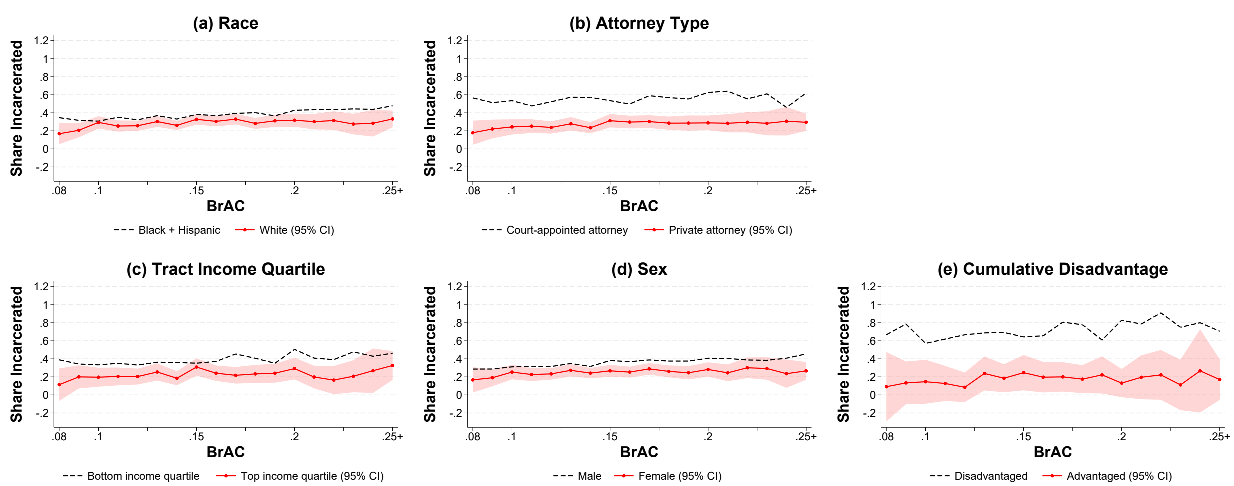
However, the disadvantages associated with race, income, gender, and representation do not merely overlap—they accumulate. The bottom-right subfigure of each panel contrasts two illustrative groups: white female defendants with private attorneys from high-income neighborhoods and Hispanic male defendants with court-appointed counsel from low-income neighborhoods. Just above the legal limit, the advantaged group has about a 70 percent likelihood of dismissal, roughly twice that of the disadvantaged group. An advantaged defendant with a BrAC of 0.25 faces higher odds of dismissal than a disadvantaged defendant with a BrAC of 0.10. The same pattern appears for incarceration: women in the advantaged group face rates below or near 20 percent across the BrAC distribution, while men in the disadvantaged group exceed 60 percent. These patterns show that each source of advantage independently improves outcomes, and that

Figure 2: Probability of Dismissal and Incarceration by Defendant Characteristics

Panel A. Dismissal



Panel B. Incarceration



Notes: This figure shows dismissal and incarceration rates of defendants within the indicated group as a function of BrAC. In sub-figure (a) of each panel, the comparison is between white defendants and Black and Hispanic defendants pooled. In sub-figure (c) of each panel, the groups are defendants from census tracts in the top versus bottom within-sample quartile of median household income. In sub-figure (e) of each panel, the advantaged group consists of white female defendants with private attorneys living in neighborhoods in the top half of the within-sample distribution of census tract median household income; the disadvantaged group consists of Hispanic male defendants with court-appointed attorneys living in neighborhoods in the bottom half. Shaded bands show 95% confidence intervals for the rate plotted as a solid line, with width equal to 1.96 times the standard error of the within-BrAC gap (estimated by regressing the outcome on a group indicator separately at each BrAC value, rounded to 0.01). The gap between groups is statistically significant where the dashed line falls outside the shaded band.

the effects accumulate to produce stark and systematic inequities.

To fully characterize the multivariate relationships between defendant characteristics and case outcomes, we estimate the following regression model:

$$Y_{i,b,d,t,c,a} = \beta X_i + \omega_b + \mu_d + \gamma_t + \phi_c + \chi_a + \epsilon_{i,b,d,t,c,a} \quad (1)$$

where Y is the case outcome for defendant i (dismissal, probation, or incarceration), and X is a set of defendant or case characteristics (e.g., race, sex, etc.). The specification includes BrAC fixed effects (ω_b) to compare defendants at identical levels of intoxication, along with day-of-week (μ_d) and month-year (γ_t) controls. We test robustness to adding courtroom-by-month-year fixed effects (ϕ_c) and arresting officer fixed effects (χ_a).

Table 2 shows the results. For dismissals, the disparities mirror the univariate patterns in Figure 2. Black and Hispanic defendants are less likely to have their cases dismissed than white defendants. Having a private attorney is positively associated with dismissal. Finally, dismissal rates increase along the tract income distribution.[11] All of these coefficients are estimated jointly and conditional on one's level of intoxication. This implies, for example, that the differences for Hispanic and white defendants do not reflect differences in sex, access to private attorney, or economic resources, nor are they due to differences in underlying intoxication.[12] With the exception of the difference by sex, we find broadly similar patterns when focusing on outright dismissals (see Table A.5).

Column 2 of Table 2 shows that the results are robust to including courtroom-by-month-year fixed effects and arresting officer fixed effects. The courtroom to which one's case is assigned is a crucial determinant of the outcome because in a given month-year courtrooms are staffed by a fixed set of judges and prosecutors (Mueller-Smith, 2015). Arresting officers also play an important role in the outcome; an officer's failure to adhere

[11] The coefficients for the second through fourth quartiles are all statistically distinguishable from the first quartile. Testing the equality of coefficients for the fourth and third quartiles yields a p-value = 0.0007 and testing the equality of the coefficients for the third and second quartiles yields a p-value = 0.2648.

[12] Recall that we restrict the main sample to citizens. See Appendix Table A.4 for joint regression results including non-citizens.

Table 2: Jointly Estimated Group Disparities

	Dismissed		Probation		Incarcerated	
	(1)	(2)	(3)	(4)	(5)	(6)
Hispanic	–0.016*	–0.019*	–0.056***	–0.052***	0.071***	0.071***
	(0.008)	(0.010)	(0.010)	(0.013)	(0.010)	(0.012)
Black	–0.032***	–0.042***	0.040***	0.047***	–0.009	–0.005
	(0.010)	(0.013)	(0.013)	(0.017)	(0.012)	(0.015)
Asian	0.030	0.035	–0.039	–0.050	0.009	0.015
	(0.024)	(0.029)	(0.027)	(0.035)	(0.025)	(0.030)
Female	0.033***	0.035***	0.033***	0.019	–0.066***	–0.054***
	(0.008)	(0.010)	(0.010)	(0.012)	(0.009)	(0.011)
Age, 26–39	0.013	–0.002	–0.028***	–0.015	0.015	0.017
	(0.008)	(0.010)	(0.010)	(0.012)	(0.009)	(0.011)
Age, 40+	0.007	–0.004	–0.018	–0.009	0.011	0.013
	(0.009)	(0.012)	(0.012)	(0.015)	(0.011)	(0.013)
Private Attorney	0.150***	0.163***	0.103***	0.096***	–0.253***	–0.259***
	(0.007)	(0.010)	(0.011)	(0.014)	(0.011)	(0.013)
Tract Inc., 2nd Qntl.	0.033***	0.041***	0.030*	0.019	–0.063***	–0.061***
	(0.012)	(0.015)	(0.015)	(0.019)	(0.014)	(0.017)
Tract Inc., 3rd Qntl.	0.046***	0.052***	0.017	0.006	–0.063***	–0.059***
	(0.012)	(0.015)	(0.015)	(0.018)	(0.014)	(0.016)
Tract Inc., 4th Qntl.	0.085***	0.094***	0.008	–0.012	–0.093***	–0.082***
	(0.012)	(0.016)	(0.015)	(0.019)	(0.014)	(0.017)
Tract Inc., Missing	0.022	0.023	–0.078***	–0.089***	0.056***	0.066***
	(0.014)	(0.018)	(0.017)	(0.021)	(0.017)	(0.020)
Dep. Var. Mean	0.218	0.218	0.453	0.453	0.329	0.329
R-squared	0.145	0.353	0.048	0.295	0.117	0.367
Observations	12,888	11,032	12,888	11,032	12,888	11,032
Court-Month-Year FEs	No	Yes	No	Yes	No	Yes
Arresting Officer FEs	No	Yes	No	Yes	No	Yes

Notes: Entries display coefficients and standard errors from regressions of case outcomes on defendant and case characteristics. Columns 1 and 2 present results on case dismissal, columns 3 and 4 on probation, and columns 5 and 6 on incarceration. Odd-numbered columns include BrAC fixed effects, day-of-week fixed effects, month-year fixed effects, and a linear control for the time of day the breath test occurred. Even-numbered columns include those variables and courtroom-month-year fixed effects and arresting officer fixed effects. We lose observations in these more detailed specifications because some arresting officers in our data only have one arrest. The following categories are omitted to avoid perfect collinearity with the included regressors: non-Hispanic whites, males, individuals aged 21-25 years old, individuals with courtappointed attorneys, and individuals from the first quartile of tract income. ***, ** and * denote statistical significance at the 1%, 5%, and 10% levels.

to legal standards or inability to appear in court can compromise the prosecution's case. Yet, when we control for these two major factors, the estimated disparities are virtually unchanged.

Next, we examine differences in probation and incarceration by defendant characteristics. In general, we find disparities that move in the same direction in terms of punishments that trade off swift, harsh sanctions versus rehabilitative approaches that, while less punitive, may entail significant costs in terms of time, fines, and cognitive load. For example, Hispanic defendants are less likely to have their cases dismissed than white defendants, and similarly, are less likely to have a sentence of probation but more likely to have a sentence of incarceration. This monotonic shift from dismissal and probation into incarceration or vice versa characterizes nearly all groups we consider. The sole exception is Black defendants, who are less likely to have their cases dismissed than white defendants but more likely to receive probation. As before, all results are robust to the inclusion of courtroom-by-month-year and arresting officer fixed effects.

To summarize, we show that defendant characteristics are strongly predictive of judicial outcomes, conditional on an objective measure of offense severity (BrAC). Additionally, race, ethnicity, attorney type, gender, and neighborhood median income all have independent predictive power for judicial outcomes even after conditioning on BrAC.

7 Threats to Comparability

We now consider the possibility that the gaps we document could be explained by a failure to construct comparisons among people that share all legally-relevant factors.

So far, we have not discussed selection into the breath testing or the set of complaint charges we use as our sample inclusion criteria. Police have a great deal of discretion over who to stop and breath test and how to write their report of the incident, while prosecutors have discretion over what charges to bring. Based on existing work (Grogger and Ridgeway, 2006; Feigenberg and Miller, 2025; Aggarwal et al., 2025), it appears that police are, on average, more likely to stop and conduct searches of lower-income people and people from racially disadvantaged groups. Work on federal prosecutors suggests that prosecutors tend to charge Black defendants with more serious offenses than similarly situated white defendants (Rehavi and Starr, 2014; Tuttle, 2025). If these patterns are mirrored in our setting, then it is possible that people from advantaged groups must engage in worse behavior to enter our sample of breath tests than those from disadvantaged groups, and that advantaged drivers facing the lowest-level DWI charges have, on average, worse underlying conduct (e.g., aggravating factors that are not charged). If these differences influenced case outcomes, it would result in *harsher* punishment for advantaged people, not more lenient punishment. We conclude that selection into the sample based on who gets breath tested and who gets charged without aggravating factors is likely to bias against the gaps that we find.

Another possibility that could generate disparities in case outcomes is the existence of substantive differences in driving behavior, conditional on BrAC. We do not observe the speed or recklessness of the driving leading up to each breath test. While these are not factors enumerated in the statute on DWI offenses, they may be reasonably taken into account by the prosecutors and judges handling these cases. If decision-makers index case outcomes to the dangerousness of driving behavior in each incident, and driving behavior, conditional on BrAC, varies across groups, then we would observe gaps in case outcomes. Though we do not observe the driving behavior of the people in our sample, we do observe how intoxicated they were. For a given person, impairment in coordination, reaction time, and vision all increase with BrAC. Voas et al. (2012) finds that a 0.02 increase in BrAC corresponds to a 54-116% increase in the risk of a fatal crash, reflecting the steep increase in driving dangerousness over the BrAC spectrum.

With this in mind, consider the differences in dangerousness, conditional on BrAC, that would need to exist across groups to account for the disparities we observe. For example, drivers from our intersectional disadvantaged group with a BrAC of 0.08 would need to be driving as badly as people from the advantaged group with a BrAC of 0.15, a difference in intoxication large enough to more than quadruple the risk of a fatal crash within a person (see Figure 3). Moving along the spectrum, drivers from the disadvantaged group with a BrAC of 0.1 would need to be driving more dangerously than even the most intoxicated advantaged drivers. Similarly, if we look at the incarceration rates in Figure 2, there is no BrAC for the advantaged groups that yields incarceration rates as high as those observed for members of the disadvantaged groups with a BrAC of 0.08. We conclude that it is implausible that differences in driving could account for the observed outcome gaps.

8 Sources of Disparities

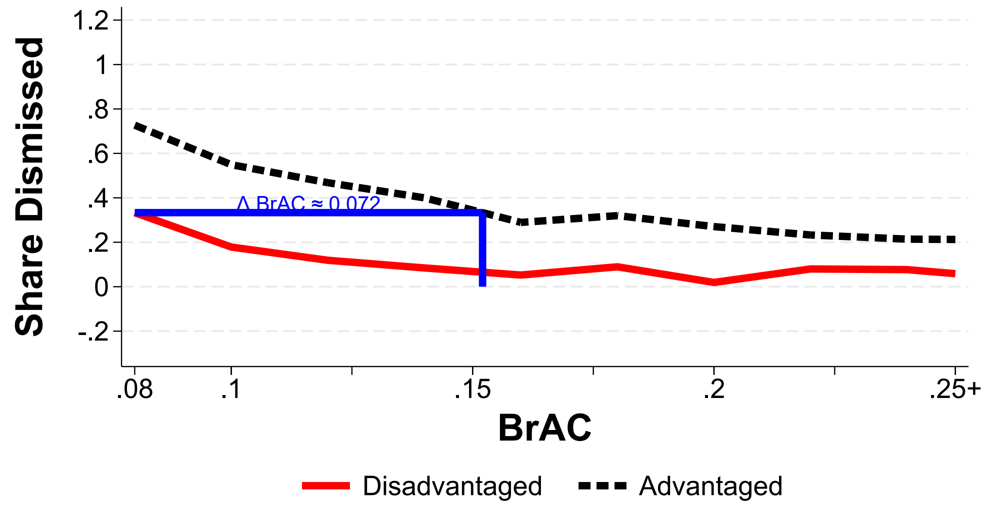
Differences in outcomes for cases with the same legally-relevant factors could arise from one or both of two sources. First, defendants from different backgrounds may face different constraints in financial resources, employment flexibility, or access to legal counsel that shape the choices available to them and the decisions they make, even when the legal options are nominally the same. Second, police, prosecutors, and judges may treat defendants differently based on group characteristics, whether through taste-based preferences, beliefs about risk, or other forms of differential treatment. Because we observe case outcomes but not the offers made to defendants, we cannot fully disentangle these two channels. Instead, we present evidence consistent with both channels, suggesting that they operate simultaneously.

8.1 Differences in Pre-existing Disadvantage

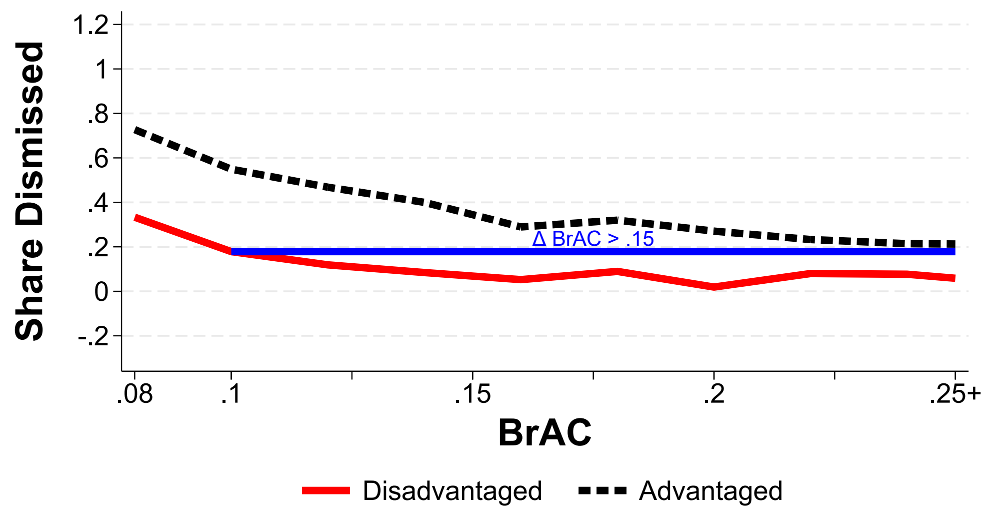
Defense attorney websites indicate that defendants can make deliberate choices among available punishment options. Perhaps, the first tradeoff lies in the choice to acquire a private attorney (Wilder Law Firm, 2025). While we do not have exogenous variation in lawyer type, our descriptive results are consistent with the possibility that hiring a pri-

Figure 3: BrACs that Equalize Dismissal Rates Across Intersectional Groups

Panel A. Horizontal Gap at BrAC = 0.08



Panel B. Horizontal Gap at BrAC = 0.10



Notes: This figure shows the dismissal rate defendants in our intersectional disadvantaged versus advantaged groups face as a function of BrAC. Panel A displays the BrAC level in the advantaged group that has the same dismissal rate as someone with a 0.08 BrAC in the disadvantaged group. Panel B highlights that no BrAC level in the advantaged group has the same dismissal rate as someone with a 0.10 BrAC in the disadvantaged group.

vate attorney substantially increases the likelihood of getting a favorable case outcome, including an outright dismissal. However, the benefits of a private attorney come with a hefty price tag. The Wilder Law Firm estimates these costs can range from $2,000 to $10,000 (Wilder Law Firm, 2025). In this way, having a private attorney is the one characteristic we consider that may both reflect defendant choices and unlock different offers.

Even holding attorney type fixed, the Wilder law firm notes defendants face choices and emphasizes that "each case is unique" and that the firm works with clients to pursue sentencing alternatives such as probation, DWI court, or alcohol education programs (Wilder Law Firm, 2025). Similarly, the Doug Murphy Law Firm describes probation as potentially burdensome, noting that it often entails hundreds of dollars in monthly costs, restrictions on alcohol consumption, mandatory classes, community service, and the risk of jail time upon noncompliance. The site makes clear that for some clients, "the lengthy requirements of the probation system are not worth it," and that a short jail sentence, while more stigmatized, may be the more manageable option (Doug Murphy Law Firm, 2025).

These descriptions suggest that part of the observed disparity in outcomes may reflect rational selection across a shared menu of sanctions. Defendants with greater systemic disadvantage due to limited financial resources, unstable employment, or inflexible schedules may reasonably prefer jail over probation, even when both options are available. For those detained pretrial, the incentive to plead guilty quickly is especially strong: a guilty plea may yield credit for time served and immediate release, even if the conviction carries longer-term costs (Leslie and Pope, 2017; Dobbie et al., 2018; Stevenson, 2018). In this way, economic constraints can push disadvantaged defendants toward incarceration over supervision.

The presence of this tradeoff between supervision and incarceration is supported by qualitative evidence. A 2010 *Texas Tribune* investigation found that many first-time DWI defendants in Harris County chose jail time rather than probation because of the financial and administrative burdens of community supervision, including monthly fees, alcohol monitoring, and frequent check-ins (Grissom, 2010). Consistent with this, Table 1 shows that incarceration is associated with shorter jail sentences but significantly lower fines than probation. Defendants who serve time pay less in fines than those receiving probation, and additional costs such as ignition interlock devices or driving school are rarely imposed on the incarcerated. These patterns imply that incarceration is the less financially costly option, while probation and diversion exchange money and supervision for reduced jail exposure and a cleaner record.

The same tradeoff appears within the set of incarcerated people. Figure A.10 shows that among defendants sentenced to jail, those with higher fines tend to serve shorter sentences—a relationship that persists even after controlling for BrAC and defendant characteristics. This inverse relationship fits a bargaining framework in which prosecutors and defendants trade time for money. Defendants with private attorneys typically pay higher fines and serve shorter sentences, suggesting that their lawyers do not necessarily help them achieve more leniency across all dimensions, but rather help them select outcomes that better align with their preferences or constraints.

Finally, the direction of disparities across demographic groups is consistent with defendants making tradeoffs. Defendants from low-income neighborhoods or without private attorneys—those least able to absorb financial penalties—are significantly more likely to receive incarceration. This pattern is consistent with rational sorting across a shared menu: when punishment options implicitly trade financial cost for reduced jail time, resource-constrained defendants may favor the lower-cost, higher-incarceration outcome. It is also possible that attorneys or prosecutors anticipate these constraints and are less likely to extend more demanding, financially costly alternatives to such defendants.

8.2 The Role of System Actors

We observe the courtroom in which each case is handled. In this setting, a courtroom consists of a fixed judge and team of prosecutors, so that the tendencies of these actors are bundled together and cannot be easily disentangled (Mueller-Smith, 2015). Since cases are randomly assigned to courtrooms, there is no reason to expect that defendant preferences over outcomes differ systematically across them, conditional on observable characteristics. Any variation in average outcomes across courtrooms therefore reflects meaningful differences in the implicit menu of options offered by different judges and prosecutors.

To quantify this variation, we estimate courtroom-by-year fixed effects from regressions of dismissal and incarceration indicators on BrAC fixed effects. Because these estimates include sampling noise, we use a split-sample approach: within each courtroom–year, we randomly divide cases in half, estimate leniency separately in each split, and compute the covariance of the two estimates. This method-of-moments approach isolates the latent variance in courtroom leniency from estimation noise. We find that a courtroom one standard deviation stricter than average is associated with a 10 percentage point reduction in dismissal probability and a 10.8 percentage point increase in incarceration probability—confirming that a substantial share of variation in criminal justice outcomes stems from prosecutorial and judicial behavior outside the control of defendants themselves.

Moreover, courtroom leniency interacts with defendant characteristics in a striking way. Figure A.8 shows that the largest gaps in dismissal rates between advantaged and disadvantaged defendants occur in the most lenient courtrooms. For example, the gap in dismissal rates between defendants with and without private attorneys exceeds 25 percentage points in the highest-dismissal courtrooms, and shrinks to nearly zero in the strictest ones; the regression coefficient relating courtroom leniency to the dismissal gap is -1.004 and highly significant. Results are similar, though somewhat weaker, for other dimensions of advantage. The pattern for incarceration, shown in Figure A.9, is somewhat less stark. For most defendant characteristics—including race and neighborhood income—gaps in incarceration rates are largely unrelated to overall courtroom incarceration rates. The exception is attorney type: defendants without private representation face substantially higher incarceration rates relative to those with private attorneys in courtrooms where incarceration is comparatively rare, echoing the dismissal pattern.

Two interpretations are consistent with these patterns. First, lenient courtrooms may extend favorable treatment selectively to advantaged defendants while withholding it from disadvantaged ones. Second, all defendants within a courtroom may nominally face the same menu of options, but advantaged defendants—by virtue of private counsel, financial resources, or schedule flexibility—are better positioned to capitalize on lenient offers. Under either interpretation, the result is the same: the benefits of judicial leniency accrue disproportionately to those who are already advantaged, amplifying rather than ameliorating pre-existing inequities. In short, courtroom leniency tends to magnify inequality rather than mitigate it. This raises a natural question: could such differential treatment be defended on efficiency grounds?

For differential treatment to have any utilitarian justification, it would need to reflect statistical discrimination based on correct beliefs. For example, if incarceration reduced recidivism, prosecutors might attempt to allocate scarce jail capacity to the riskiest defendants to maximize future public safety. This would require decision-makers to correctly distinguish between people with higher and lower risk of future bad behavior and assign differential treatment in a way that actually mitigated that risk. In our sample, gaps in future recidivism across groups are relatively small, which could mean either that groups do not differ meaningfully in their future riskiness or that court actors are successfully engaging in statistical discrimination.

To distinguish between these possibilities, we estimate whether dismissal or incarceration affect the probability of future criminal charges or future DWI charges, using courtroom fixed effects as instruments for treatment. The results are shown in Table 3. While the sample size limits precision and the presence of multiple treatment categories complicates inference in a judge leniency design, the point estimates are inconsistent with rational statistical discrimination. If anything, incarceration *increases* the probability of recidivism and dismissal *decreases* it, consistent with a growing body of evidence on the criminogenic effects of punitive sanctions (Mueller-Smith and Schnepel, 2020; Agan et al., 2023). This makes it difficult to rationalize differential treatment on efficiency grounds. If system actors are making different offers to different defendants, the evidence points toward statistical discrimination based on incorrect beliefs or taste-based discrimination rather than accurate risk-targeting. It bears emphasis that we see similar cross-group disparities in outright dismissals, an outcome that presumably dominates all others from a defendant's perspective, making it difficult to attribute these gaps entirely to differences in defendant choices over a shared menu.

Table 3: Relationship between Punishment and Recidivism

	Any New Charge		New DWI Charge	
	(1)	(2)	(3)	(4)
Dismissed	–0.029		–0.037	
	(0.079)		(0.059)	
Incarcerated		0.008		0.033
		(0.037)		(0.028)
Dependent Var. Mean	0.179	0.179	0.087	0.087
Observations	12,786	12,786	12,786	12,786

Notes: Entries display coefficients and standard errors from UJIVE estimates of the effect of the defendant's punishment on their likelihood of recidivism. Columns 1 and 2 present results for the likelihood the defendant has any new charge within five years, and columns 3 and 4 present results for likelihood of a new DWI charge within five years. Odd-numbered columns estimate the effect of case dismissal and even-numbered columns estimate the effect of incarceration. ***, ** and * denote statistical significance at the 1%, 5%, and 10% levels.

As discussed, we cannot parse how much of the observed gaps arise from pre-existing disadvantage shaping defendant choices versus differential treatment by system actors.

However, we find suggestive evidence that the differences in outcomes cannot be rationalized on the basis of accurate statistical discrimination. Instead we conclude that the patterns we observe likely result from a combination of inaccurate statistical discrimination, taste-based discrimination, and differences in pre-existing disadvantage; in short, extralegal factors unrelated to the facts of the case substantively shape criminal case outcomes.

9 Conclusion

This paper documents substantial disparities in DWI case outcomes among defendants who are similarly situated under the law. By focusing on first-time offenders without aggravating circumstances and conditioning on BrAC, an objective measure of legal guilt, we isolate differences in punishment along four dimensions: race/ethnicity, gender, use of a private attorney, and neighborhood income. Each dimension matters conditional on the others, and the gaps aggregate cumulatively. The magnitudes are too large to be plausibly explained by unobserved differences in driving behavior. Equal guilt does not translate into equal punishment.

Our findings point to two complementary channels through which extralegal factors shape case outcomes. First, disparities may reflect differential treatment by prosecutors or judges in charging, plea bargaining, or sentencing decisions. Second, they may arise because defendants differ in their ability to navigate the same set of formal options: probation and diversion are often more financially costly than incarceration, requiring upfront fees, stable employment, reliable transportation, and flexible schedules. For defendants facing financial hardship or housing instability, incarceration may be the only feasible resolution. In this way, pre-existing disadvantage constrains which outcomes are practically available, producing disparities even absent differential treatment by system actors.

Importantly, these channels are not mutually exclusive, and both carry long-term consequences. A growing body of research shows that criminal case outcomes influence future criminal justice contact, labor market attachment, and recidivism. If disadvantaged defendants systematically end up with harsher sanctions—whether because of how they are treated or the constraints they face—initial disparities may compound over time. What appears to be a short-term sentencing difference can become a driver of persistent inequality. Strikingly, gaps in case outcomes are largest in the most lenient courtrooms. Systems designed to offer flexibility and rehabilitative alternatives may inadvertently magnify inequality when access to those alternatives depends on private resources and discretionary treatment by court actors.

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Online Appendix A: Supplementary Tables and Figures

Appendix Table A.1: Summary Statistics

	All Cases	Dismissed	Probation	Incarceration
Case Outcomes				
Dismissed	0.218	1	0	0
	(0.413)	-	-	-
Probation	0.453	0	1	0
	(0.498)	-	-	-
Incarceration	0.329	0	0	1
	(0.470)	-	-	-
Defendant Characteristics				
Hispanic	0.326	0.289	0.292	0.396
	(0.469)	(0.453)	(0.455)	(0.489)
White	0.496	0.536	0.518	0.438
	(0.500)	(0.499)	(0.500)	(0.496)
Black	0.148	0.133	0.161	0.140
	(0.355)	(0.339)	(0.368)	(0.347)
Asian	0.025	0.034	0.024	0.021
	(0.157)	(0.181)	(0.154)	(0.145)
Female	0.267	0.305	0.293	0.207
	(0.442)	(0.460)	(0.455)	(0.405)
Age	33.259	32.917	33.232	33.524
	(11.173)	(10.903)	(11.154)	(11.370)
Private Attorney	0.801	0.934	0.836	0.663
	(0.400)	(0.249)	(0.370)	(0.473)
Med. Household Inc. (Tract-Level, $k)	58.097	62.265	58.435	53.891
	(26.460)	(27.340)	(25.791)	(26.097)
BrAC	0.157	0.142	0.161	0.160
	(0.044)	(0.040)	(0.044)	(0.044)
Observations	12,888	2,811	5,836	4,241

Notes: Entries display means and standard deviations for case outcomes and defendant characteristics. Column 1 reports these statistics for the full sample. Columns 2-4 report these statistics, respectively, for cases that end in dismissal, probation, or incarceration. We observe tract-level median income for 75% of our sample ($n = 9,571$); the mean and standard deviation for that variable are thus based on those observations.

Appendix Table A.2: Sentencing Outcomes by Defendant Type

Defendant Type	**Proportion of Defendants**			***n***
	Dismissed	Probation	Incarceration	
Private attorney				
Asian female	0.333	0.417	0.250	60
Asian male	0.303	0.447	0.250	228
White female	0.287	0.502	0.210	1,816
Hispanic female	0.287	0.461	0.253	558
White male	0.264	0.468	0.268	3,450
Black female	0.249	0.502	0.249	414
Black male	0.233	0.503	0.263	1,025
Hispanic male	0.213	0.453	0.334	2,710
Court-appointed attorney				
Hispanic female	0.136	0.466	0.398	103
Hispanic male	0.074	0.210	0.717	829
White male	0.067	0.400	0.533	777
Black male	0.067	0.445	0.488	344
Black female	0.065	0.520	0.415	123
White female	0.058	0.541	0.401	344

Notes: Entries display the share of defendants by disposition type within demographic-by-attorney cells for all cells with greater than 50 cases. We organize the table into two main groups: (1) defendants who have a private attorney and (2) defendants who do not have a private attorney. Within each group, we report the share of cases that end in dismissal, probation, or incarceration by race/ethnicity and sex. The last column reports the number of cases in each cell.

Appendix Table A.3: Jointly Estimated Group Disparities, Strict Prior Conviction Filter

	Dismissed		Probation		Incarcerated	
	(1)	(2)	(3)	(4)	(5)	(6)
Hispanic	–0.032***	–0.033***	–0.051***	–0.045***	0.083***	0.078***
	(0.009)	(0.012)	(0.011)	(0.014)	(0.010)	(0.013)
Black	–0.030***	–0.039**	0.042***	0.045**	–0.012	–0.006
	(0.011)	(0.015)	(0.014)	(0.019)	(0.013)	(0.016)
Asian	0.021	0.018	–0.040	–0.054	0.019	0.035
	(0.025)	(0.031)	(0.029)	(0.037)	(0.026)	(0.032)
Female	0.030***	0.034***	0.036***	0.018	–0.066***	–0.052***
	(0.009)	(0.011)	(0.011)	(0.014)	(0.009)	(0.011)
Age, 26–39	0.016*	–0.004	–0.037***	–0.020	0.021**	0.024**
	(0.009)	(0.011)	(0.011)	(0.014)	(0.010)	(0.012)
Age, 40+	0.005	–0.009	–0.029**	–0.009	0.024**	0.018
	(0.010)	(0.013)	(0.013)	(0.016)	(0.011)	(0.014)
Private Attorney	0.150***	0.167***	0.088***	0.087***	–0.238***	–0.254***
	(0.008)	(0.011)	(0.012)	(0.016)	(0.012)	(0.015)
Tract Inc., 2nd Qntl.	0.032**	0.034**	0.028*	0.016	–0.060***	–0.050***
	(0.014)	(0.017)	(0.017)	(0.021)	(0.015)	(0.018)
Tract Inc., 3rd Qntl.	0.048***	0.048***	0.014	0.004	–0.062***	–0.052***
	(0.013)	(0.017)	(0.016)	(0.021)	(0.015)	(0.018)
Tract Inc., 4th Qntl.	0.086***	0.087***	0.003	–0.013	–0.089***	–0.073***
	(0.013)	(0.017)	(0.016)	(0.021)	(0.015)	(0.018)
Tract Inc., Missing	0.019	0.016	–0.098***	–0.119***	0.079***	0.102***
	(0.015)	(0.020)	(0.019)	(0.024)	(0.018)	(0.022)
Dep. Var. Mean	0.233	0.233	0.451	0.451	0.315	0.315
R-squared	0.160	0.387	0.053	0.316	0.121	0.392
Observations	11,085	9,395	11,085	9,395	11,085	9,395
Court-Month-Year FEs	No	Yes	No	Yes	No	Yes
Arresting Officer FEs	No	Yes	No	Yes	No	No

Notes: Entries display coefficients and standard errors from regressions of case outcomes on defendant and case characteristics. In this table, we remove individuals who link, on the basis of name and age, to a record in the Texas-wide conviction database that occurs prior to their case in Harris County. Columns 1 and 2 present results on case dismissal, columns 3 and 4 on probation, and columns 5 and 6 on incarceration. Odd-numbered columns include BrAC fixed effects, day-of-week fixed effects, month-year fixed effects, and a linear control for the time of day the breath test occurred. Even-numbered columns include those variables and courtroom-month-year fixed effects and arresting officer fixed effects. We lose observations in these more detailed specifications because some arresting officers in our data only have one arrest. The following categories are omitted to avoid perfect collinearity with the included regressors: non-Hispanic whites, males, individuals aged 21-25 years old, individuals with court-appointed attorneys, and individuals from the first quartile of tract income. ***, ** and * denote statistical significance at the 1%, 5%, and 10% levels.

Appendix Table A.4: Jointly Estimated Group Disparities, Including Non-Citizens

	Dismissed		Probation		Incarcerated	
	(1)	(2)	(3)	(4)	(5)	(6)
Hispanic	–0.021***	–0.023***	–0.060***	–0.053***	0.082***	0.075***
	(0.007)	(0.008)	(0.009)	(0.010)	(0.008)	(0.009)
Black	–0.024**	–0.030**	0.054***	0.058***	–0.030**	–0.029**
	(0.010)	(0.012)	(0.013)	(0.015)	(0.012)	(0.014)
Asian	0.043**	0.046*	–0.017	–0.024	–0.027	–0.022
	(0.021)	(0.024)	(0.025)	(0.029)	(0.023)	(0.026)
Female	0.033***	0.034***	0.051***	0.044***	–0.085***	–0.077***
	(0.008)	(0.009)	(0.009)	(0.011)	(0.009)	(0.010)
Age, 26–39	0.008	0.005	–0.022***	–0.021**	0.014*	0.016*
	(0.006)	(0.007)	(0.008)	(0.009)	(0.007)	(0.009)
Age, 40+	0.011	0.006	0.007	0.009	–0.018**	–0.015
	(0.007)	(0.009)	(0.009)	(0.011)	(0.009)	(0.010)
Citizen	0.046***	0.047***	0.129***	0.134***	–0.175***	–0.181***
	(0.006)	(0.008)	(0.008)	(0.010)	(0.008)	(0.009)
Private Attorney	0.133***	0.142***	0.182***	0.180***	–0.315***	–0.322***
	(0.005)	(0.006)	(0.008)	(0.009)	(0.008)	(0.009)
Tract Inc., 2nd Qntl.	0.025***	0.028***	0.026**	0.029**	–0.051***	–0.056***
	(0.008)	(0.010)	(0.010)	(0.012)	(0.010)	(0.012)
Tract Inc., 3rd Qntl.	0.042***	0.040***	0.031***	0.029**	–0.073***	–0.069***
	(0.009)	(0.010)	(0.011)	(0.013)	(0.011)	(0.012)
Tract Inc., 4th Qntl.	0.087***	0.087***	0.018	0.006	–0.105***	–0.093***
	(0.010)	(0.012)	(0.012)	(0.014)	(0.011)	(0.013)
Tract Inc., Missing	0.015*	0.016	–0.066***	–0.068***	0.050***	0.052***
	(0.009)	(0.011)	(0.011)	(0.013)	(0.011)	(0.013)
Dep. Var. Mean	0.175	0.175	0.361	0.361	0.464	0.464
R-squared	0.146	0.307	0.123	0.304	0.263	0.432
Observations	18,801	16,546	18,801	16,546	18,801	16,546
Court-Month-Year FEs	No	Yes	No	Yes	No	Yes
Arresting Officer FEs	No	Yes	No	Yes	No	No

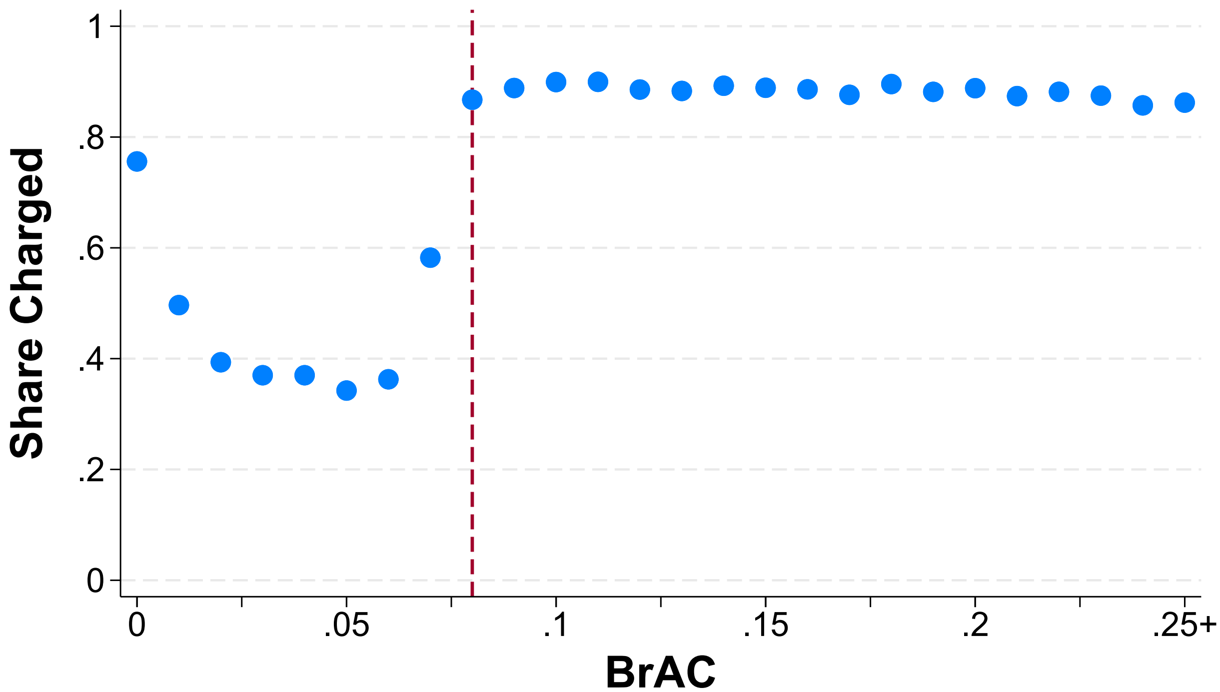
Notes: Entries display coefficients and standard errors from regressions of case outcomes on defendant and case characteristics. In this table, we include defendants who are flagged as non-U.S. citizens in the court records. Columns 1 and 2 present results on case dismissal, columns 3 and 4 on probation, and columns 5 and 6 on incarceration. Odd-numbered columns include BrAC fixed effects, day-of-week fixed effects, month-year fixed effects, and a linear control for the time of day the breath test occurred. Even-numbered columns include those variables and courtroom-month-year fixed effects and arresting officer fixed effects. We lose observations in these more detailed specifications because some arresting officers in our data only have one arrest. The following categories are omitted to avoid perfect collinearity with the included regressors: non-Hispanic whites, males, individuals aged 21-25 years old, non-citizens, individuals with court-appointed attorneys, and individuals from the first quartile of tract income. ***, ** and * denote statistical significance at the 1%, 5%, and 10% levels.

Appendix Table A.5: Jointly Estimated Group Disparities

	Outright Dismissal		Diversion Dismissal	
	(1)	(2)	(3)	(4)
Hispanic	–0.024***	–0.024***	0.008	0.006
	(0.005)	(0.006)	(0.007)	(0.010)
Black	–0.016**	–0.023***	–0.016*	–0.019
	(0.006)	(0.008)	(0.009)	(0.012)
Asian	0.001	0.011	0.029	0.025
	(0.014)	(0.017)	(0.022)	(0.027)
Female	–0.008*	–0.009	0.041***	0.043***
	(0.005)	(0.006)	(0.007)	(0.010)
Age, 26–39	0.006	–0.002	0.007	–0.001
	(0.005)	(0.006)	(0.007)	(0.009)
Age, 40+	–0.003	–0.018***	0.011	0.014
	(0.005)	(0.007)	(0.008)	(0.011)
Private Attorney	0.035***	0.041***	0.115***	0.122***
	(0.004)	(0.006)	(0.006)	(0.009)
Tract Inc., 2nd Qntl.	0.009	0.013	0.024**	0.028**
	(0.007)	(0.009)	(0.011)	(0.014)
Tract Inc., 3rd Qntl.	0.009	0.018**	0.036***	0.034**
	(0.007)	(0.008)	(0.011)	(0.014)
Tract Inc., 4th Qntl.	0.027***	0.033***	0.057***	0.061***
	(0.007)	(0.009)	(0.011)	(0.014)
Tract Inc., Missing	0.023***	0.022**	–0.001	0.000
	(0.008)	(0.010)	(0.012)	(0.016)
Dep. Var. Mean	0.060	0.060	0.158	0.158
R-squared	0.096	0.322	0.108	0.316
Observations	12,888	11,032	12,888	11,032
Court-Month-Year FEs	No	Yes	No	Yes
Arresting Officer FEs	No	Yes	No	Yes

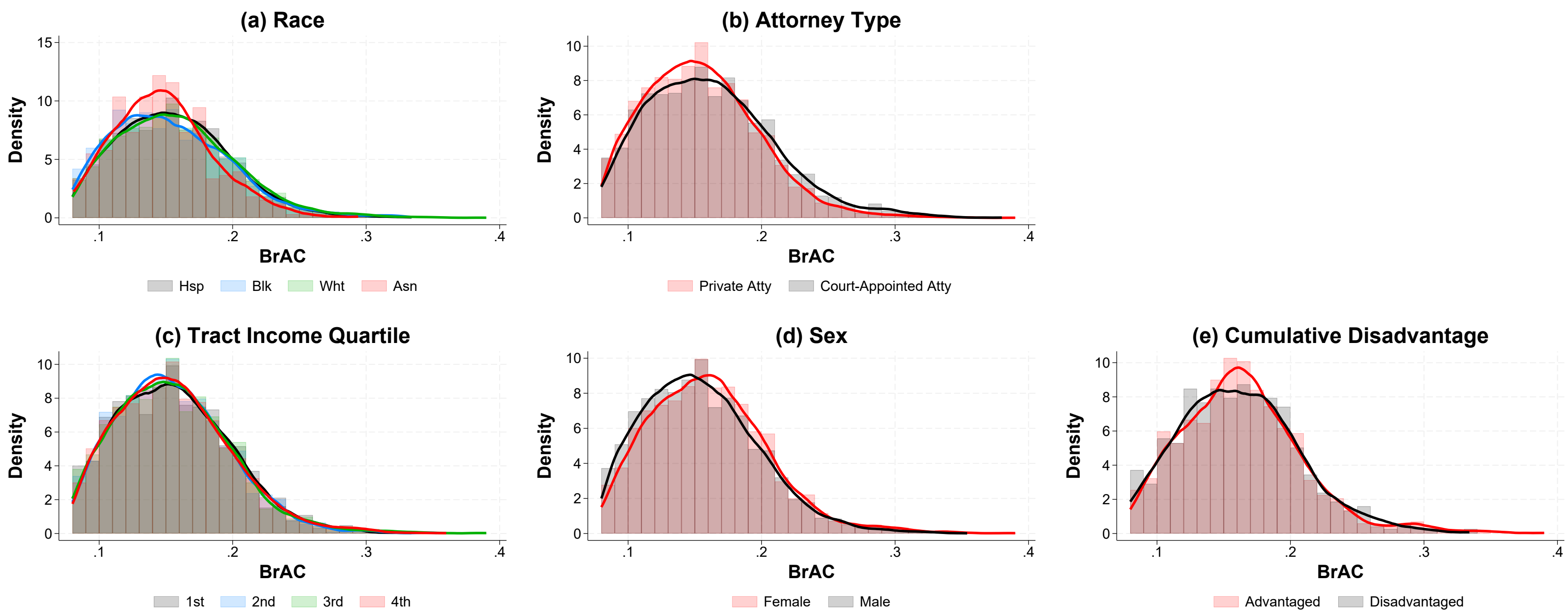
Notes: Entries display coefficients and standard errors from regressions of case outcomes on defendant and case characteristics. Columns 1 and 2 present results on outright dismissals, and columns 3 and 4 on dismissals after pre-trial diversion. Odd-numbered columns include BrAC fixed effects, day-of-week fixed effects, month-year fixed effects, and a linear control for the time of day the breath test occurred. Even-numbered columns include those variables and courtroom-month-year fixed effects and arresting officer fixed effects. We lose observations in these more detailed specifications because some arresting officers in our data only have one arrest. The following categories are omitted to avoid perfect collinearity with the included regressors: non-Hispanic whites, males, individuals aged 21-25 years old, individuals with court-appointed attorneys, and individuals from the first quartile of tract income. ***, ** and * denote statistical significance at the 1%, 5%, and 10% levels.

Appendix Figure A.1: Share of Incidents Charged with DWI by BrAC



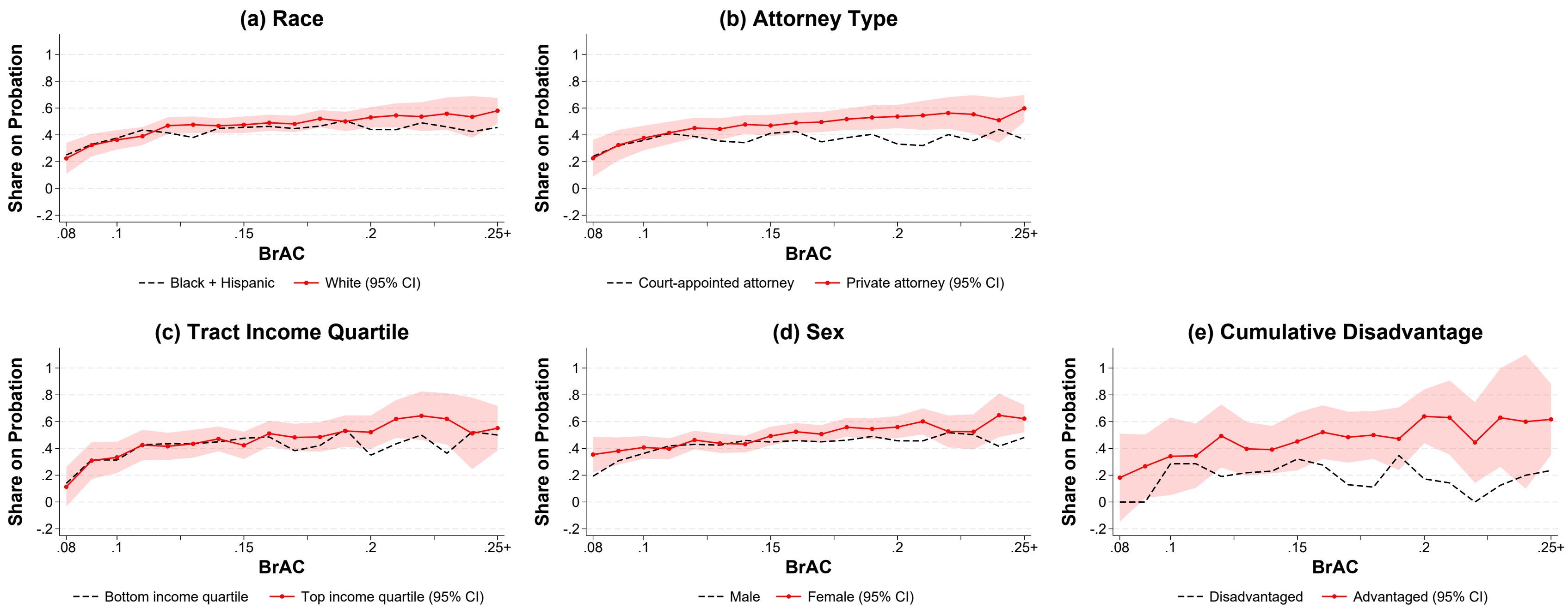
Notes: Figure displays the share of breath test incidents in Harris County that result in a charge in Harris County Courts by BrAC. The legal limit of 0.08 is marked by the red dashed line. Approximately 90% of cases above the legal limit result in a charge, and above that cutoff, the likelihood of a charge is uncorrelated with severity of the offense. Cases below the legal limit may result in a charge, and often do, because a driver can be found legally guilty of DWI if it is determined alcohol or drugs impaired their driving ability, regardless of their BrAC.

Appendix Figure A.2: BrAC Distributions by Defendant Characteristics



Notes: Figure displays the distribution of BrAC in incidents above the legal limit by defendant characteristics. Panel (a) displays these distributions by race, panel (b) by attorney type, panel (c) by quartiles of tract-level median income and panel (d) by sex. Panel (e) displays these distributions for the defendant groups we use to illustrate cumulative disadvantage: white females from the top half of the tract-income distribution and with a private attorney (in red) versus Hispanic males from the bottom half of the tract-income distribution and with a court-appointed attorney (in black).

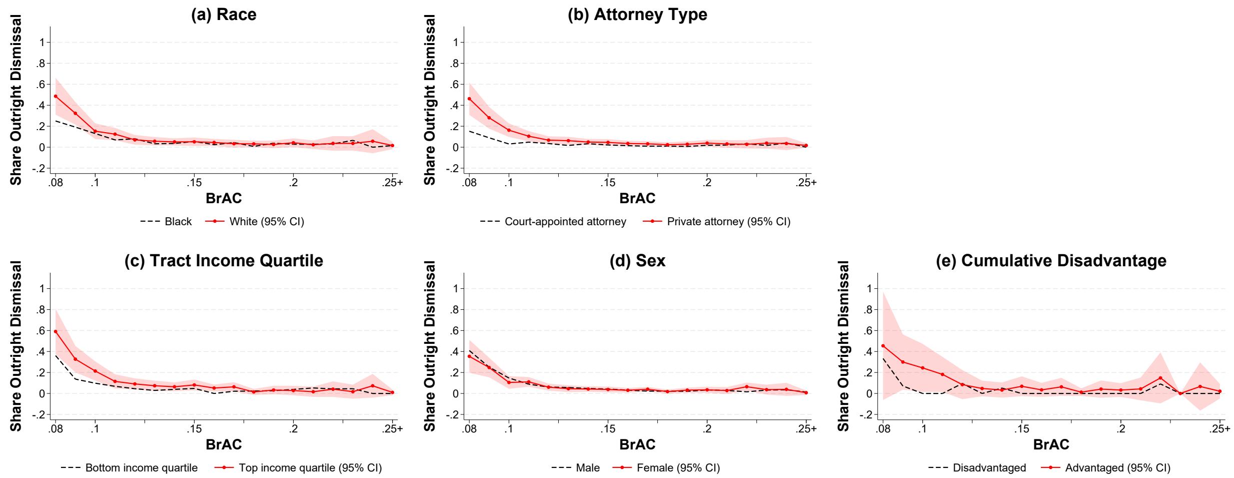
Appendix Figure A.3: Probability of Probation by Defendant Characteristics



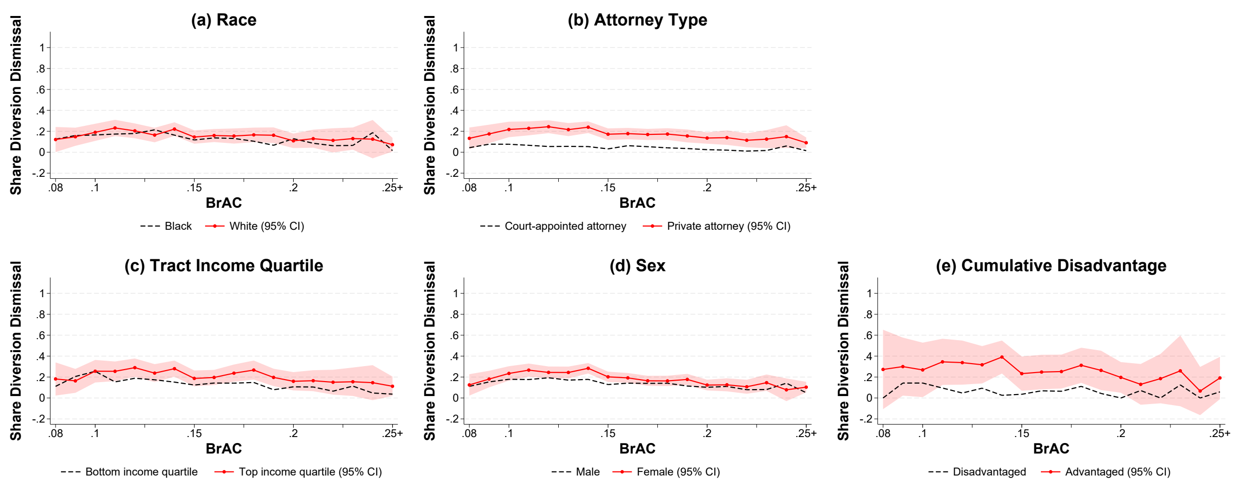
Notes: This figure shows the probation rate of defendants within the indicated group as a function of BrAC. In the bottom left panel of the figure, the groups are constructed based on the within-sample quartile of census tract median household income. In the bottom right panel, the advantaged group is constructed of defendants who are white females with private attorneys living in neighborhoods in the top half of the within-sample distribution of census tract median household income. The disadvantaged group is composed of defendants who are Hispanic males with court-appointed attorneys living in neighborhoods in the bottom half of the within-sample distribution of census tract median household income. Shaded bands show 95% confidence intervals for the rate plotted as a solid line, with width equal to 1.96 times the standard error of the within-BrAC gap (estimated by regressing the outcome on a group indicator separately at each BrAC value, rounded to 0.01). The gap between groups is statistically significant where the dashed line falls outside the shaded band.

Appendix Figure A.4: Probability of Outright Dismissal and Dismissal After Diversion by Defendant Characteristics

Panel A. Outright Dismissal

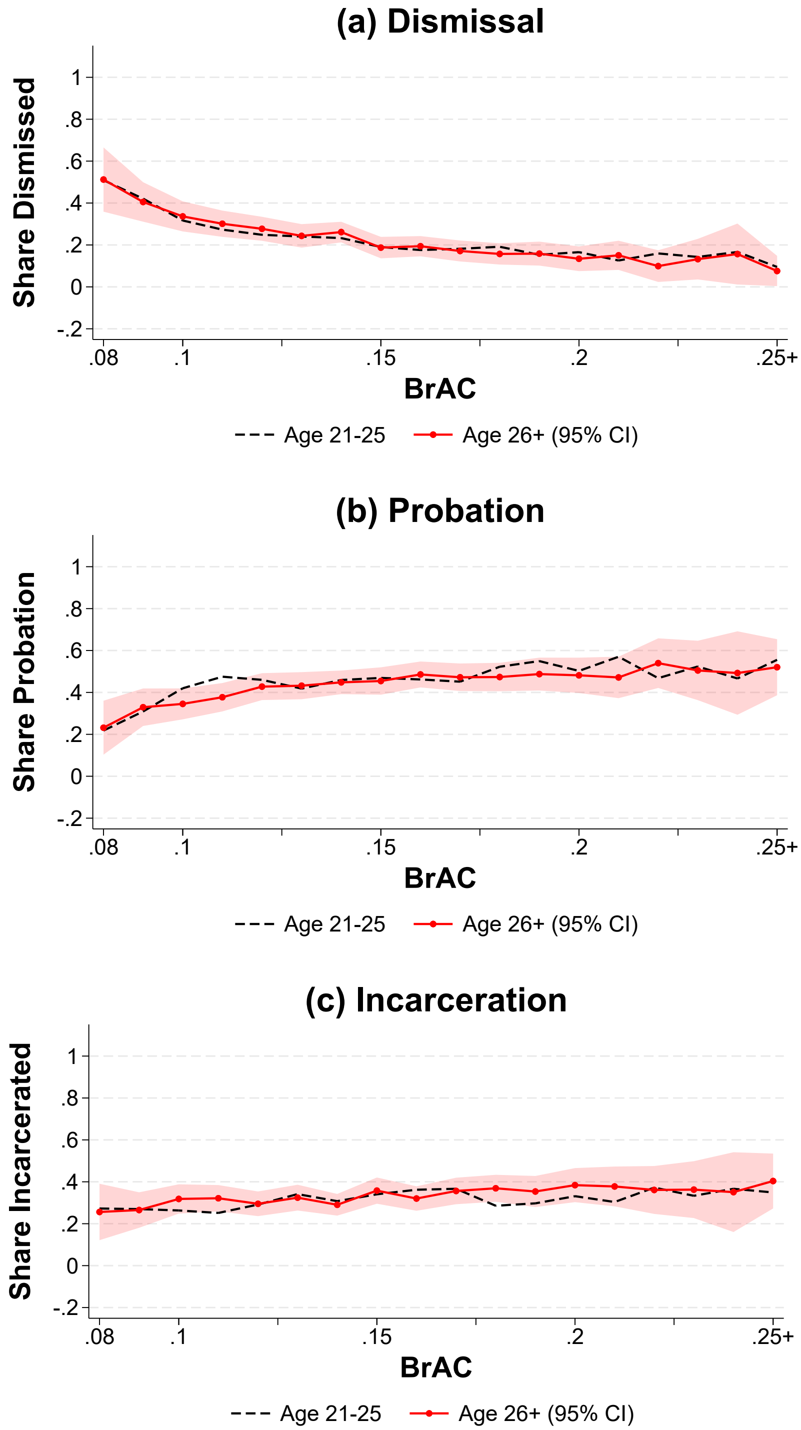


Panel B. Dismissal After Diversion



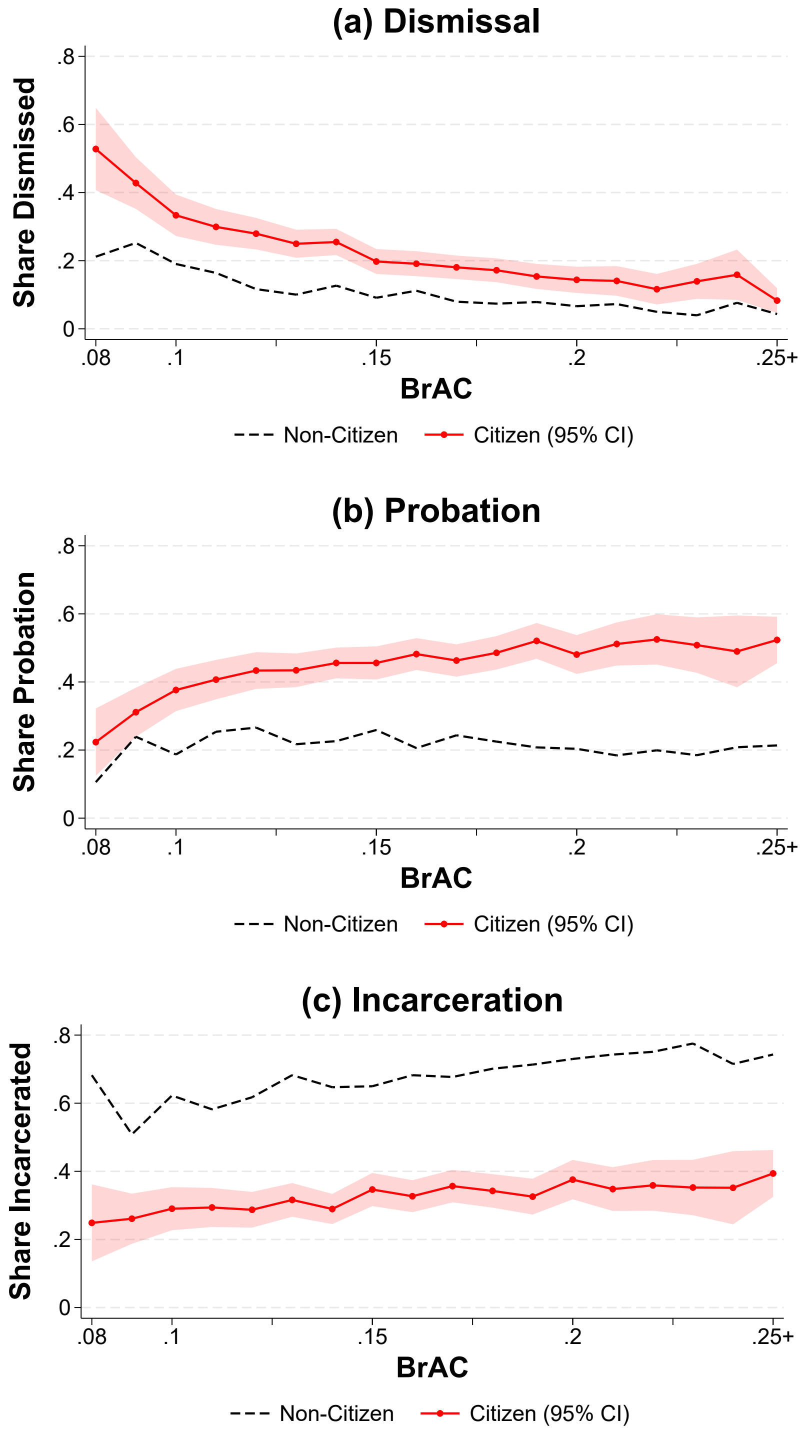
Notes: This figure shows outright dismissal and dismissal after diversion rates of defendants within the indicated group as a function of BrAC. Outright dismissal is proxied for with an indicator for having a dismissal within a year of the breath test. Dismissal after diversion is proxied for with an indicator for having a dismissal more than a year after the breath test. In sub-figure (c) of each panel, the groups are constructed based on the within-sample quartile of census tract median household income. In sub-figure (e) of each panel, the advantaged group is constructed of defendants who are white female defendants with private attorneys living in neighborhoods in the top half of the within-sample distribution of census tract median household income. The disadvantaged group is composed of defendants who are Hispanic male defendants with court-appointed attorneys living in neighborhoods in the bottom half of the within-sample distribution of census tract median household income. Shaded bands show 95% confidence intervals for the rate plotted as a solid line, with width equal to 1.96 times the standard error of the within-BrAC gap (estimated by regressing the outcome on a group indicator separately at each BrAC value, rounded to 0.01). The gap between groups is statistically significant where the dashed line falls outside the shaded band.

Appendix Figure A.5: Probability of Disposition Types by Defendant Age



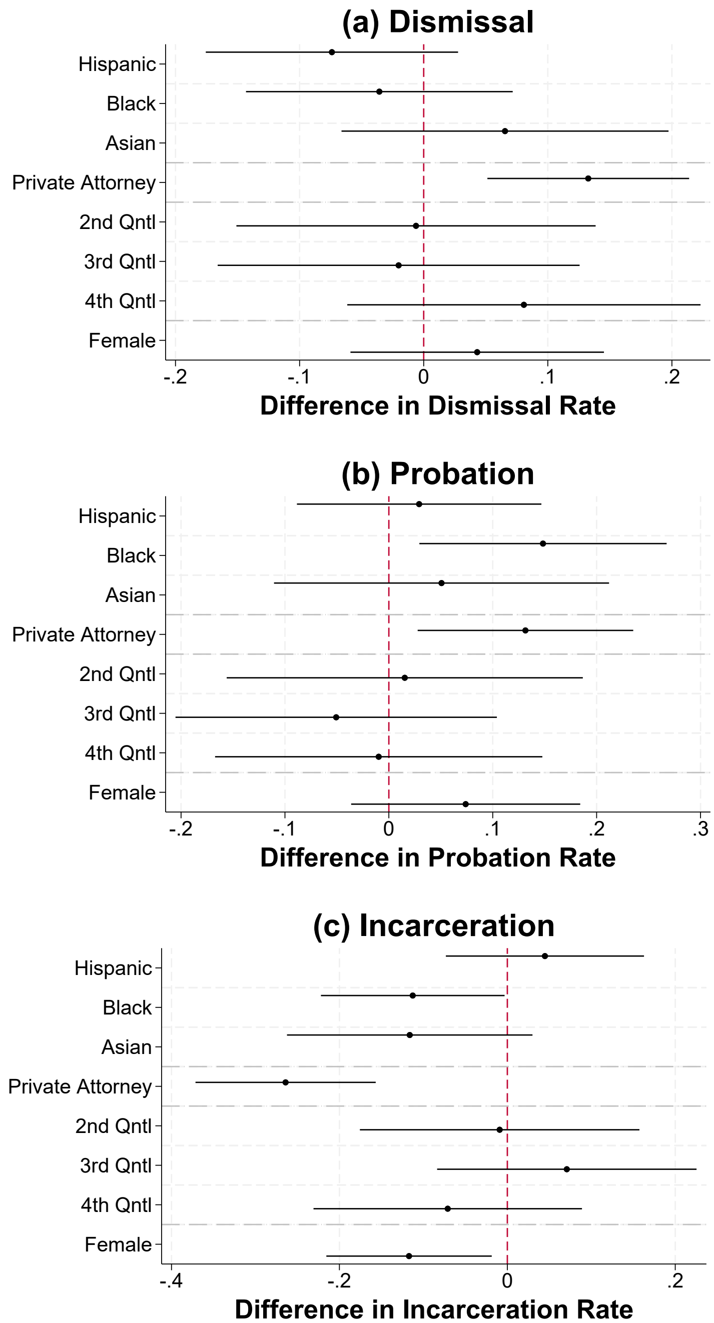
Notes: Figure displays the dismissal, probation, and incarceration rates for defendants by age group and as a function of BrAC. 21-25 year olds represent about 30% of the sample and 26 year olds and above represent about 70% of the sample. Shaded bands show 95% confidence intervals for the rate plotted as a solid line, with width equal to 1.96 times the standard error of the within-BrAC gap (estimated by regressing the outcome on a group indicator separately at each BrAC value, rounded to 0.01). The gap between groups is statistically significant where the dashed line falls outside the shaded band.

Appendix Figure A.6: Probability of Disposition Types by Defendant Citizenship



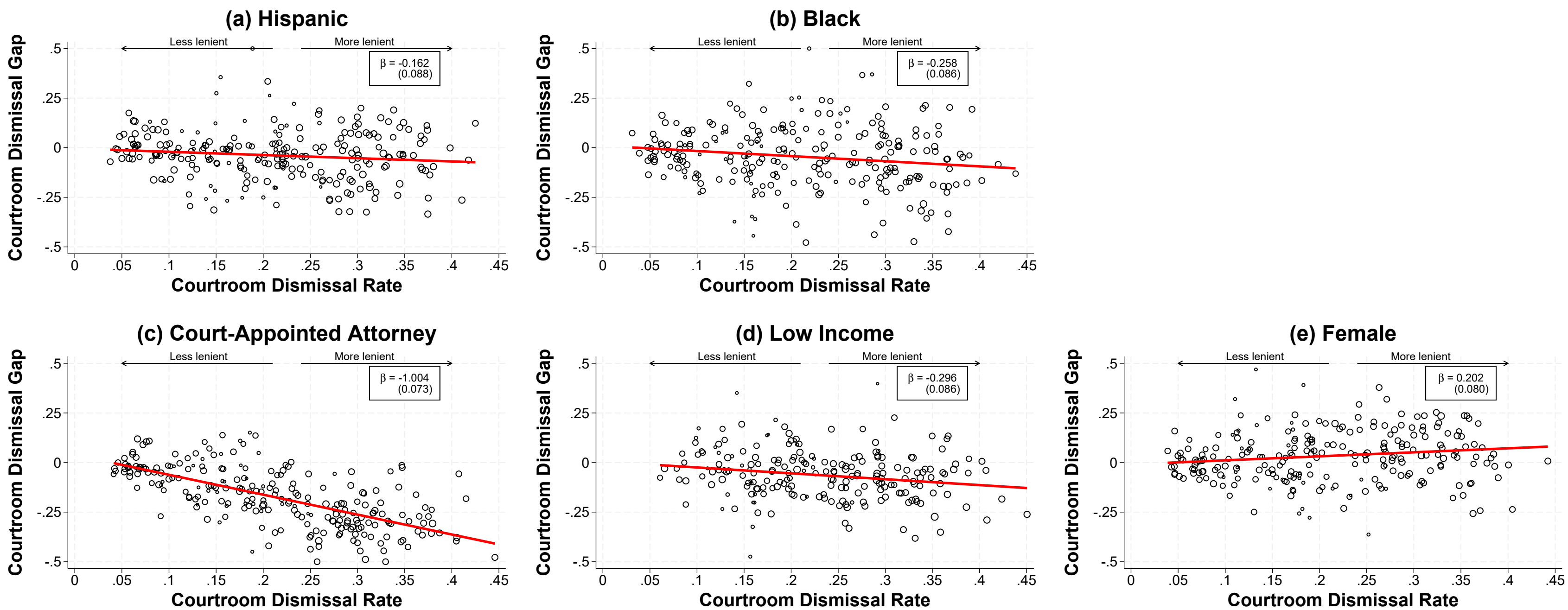
Notes: Figure displays the dismissal, probation, and incarceration rates for defendants by citizenship status and as a function of BrAC. Citizens represent about 60% of the sample, non-citizens and individuals missing citizenship information represent about 40% of the sample. Shaded bands show 95% confidence intervals for the rate plotted as a solid line, with width equal to 1.96 times the standard error of the within-BrAC gap (estimated by regressing the outcome on a group indicator separately at each BrAC value, rounded to 0.01). The gap between groups is statistically significant where the dashed line falls outside the shaded band.

Appendix Figure A.7: Raw Disparities in Outcomes for Detailed Sample



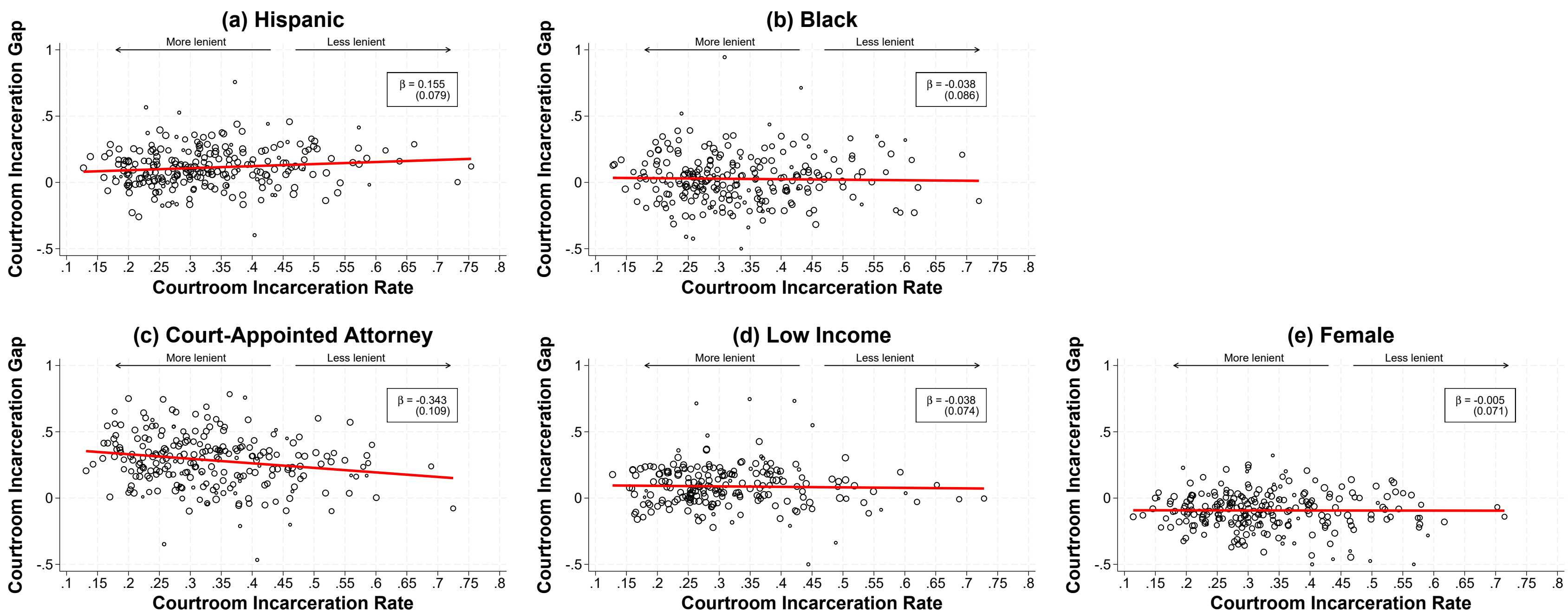
Notes: In this figure, we restrict attention to the subset of cases for which we collected information directly from case dockets, and for which we can confirm that there is no mention of prior criminal history at any point in the docket. Then, we estimate mean differences by defendant characteristics, controlling linearly for defendant age, time of day of the breath test and including BrAC, month-year, and day-of-week fixed effects. Given the small sample, we estimate these regressions separately for race, attorney type, tract-level median income, and sex. In each panel, we plot the coefficients and 95% confidence intervals from these four regressions. The top panel examines disparities in dismissal, the middle panel examines disparities in probation, and the bottom panel examines disparities in incarceration.

Appendix Figure A.8: Relationship between Courtroom Dismissal Rate and Group Disparities



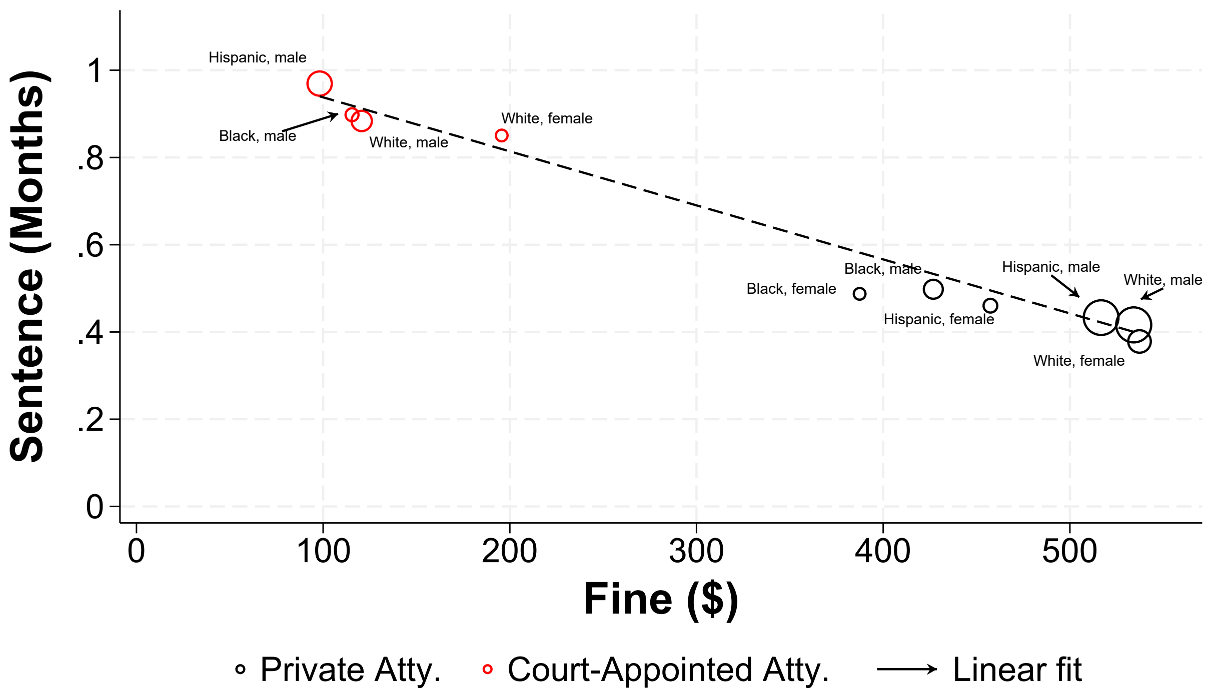
Notes: This figure shows the relationship between overall courtroom-level leniency and the courtroom-level gap in dismissal between defendants in the indicated group and an omitted reference category. Since courtroom-level gaps are mechanically related to overall generosity measures, a simple regression would be problematic. To overcome this, we use our split-sample methodology: for each courtroom–year, we calculate group gaps in one split and relate them to an empirical Bayes estimate of courtroom leniency from the other split. This design eliminates mechanical correlation and yields a clean estimate of how leniency shapes disparities. Defendants are randomly assigned into splits separately for each panel of the figure, stratified to ensure balance on the reference characteristic. Both the gaps and the leniency measures are normalized to have population means equal to the overall average. Courtroom is defined as the interaction between court ID and year, to account for turnover in prosecutors, judges, etc. over time (Mueller-Smith, 2015). In sub-figures (a) and (b), the gaps are calculated relative to white defendants. In sub-figure (c), the gap is calculated relative to defendants with a privately retained attorney. In sub-figure (d), the gap is calculated relative to defendants who live in tracts in the top two quartiles of the census tract median household income distribution. Finally, in sub-figure (e), the gap is calculated relative to male defendants. For data visualization purposes, we winsorize the gap at 0.5 and -0.5. The slope is estimated on the raw data, pre-winsorization.

Appendix Figure A.9: Relationship between Courtroom Incarceration Rate and Group Disparities



Notes: This figure shows the relationship between overall courtroom-level leniency and the courtroom-level gap in incarceration between defendants in the indicated group and an omitted reference category. Courtroom is defined as the interaction between court ID and year, to account for turnover in prosecutors, judges, etc. over time (Mueller-Smith, 2015). In sub-figures (a) and (b), the gaps are calculated relative to white defendants. In sub-figure (c), the gap is calculated relative to defendants with a privately retained attorney. In sub-figure (d), the gap is calculated relative to defendants who live in tracts in top two quartiles of the census tract median household income distribution. Finally, in sub-figure (e), the gap is calculated relative to male defendants. For data visualization purposes, we winsorize the gap at 1.0 and -0.5. The slope is estimated on the raw data, pre-winsorization.

Appendix Figure A.10: Fine and Sentence Length across Defendant Characteristics



Notes: In this figure, we restrict attention to the subset of cases that end in incarceration. Then, we construct the average incarceration length and fine amount within defendant race/ethnicity, sex, and attorney type cells. We plot these means for cells with more than 100 cases. We also plot the linear fit between average sentence length and average fine amount. The linear fit and the scatter plot markers are both weighted by the number of cases in each cell.

Online Appendix B: Data and Matching Procedure

B.1 Harris County Court Records

We rely on detailed records from Harris County District Courts to examine defendant characteristics and case outcomes. These data were obtained via a public records request, and they cover all cases disposed in Harris County District Courts from 1990 to the beginning of 2021. These records contain case details, such as the charged offense and the type of attorney representing the defendant (i.e., private versus court-appointed); case outcomes, such as case disposition (i.e., dismissal, probation, or incarceration), sentence length, and fine amount; and defendant characteristics, such as race/ethnicity, sex, citizenship, and full residential address.

Crucially for our purposes, the records also include the full name of the defendant, their exact date of birth, and the date on which the case was filed. We use these details in our matching procedure outlined below. We also leverage the long time-span of the data to construct detailed criminal histories for each defendant. For example, for cases filed in 2004, we can observe all prior charges in Harris County District Courts over the last fourteen years. Finally, we supplement this dataset with data on tract-level median household income from the American Community Survey (ACS). We geocode and assign a tract based on addresses from the court records. We then link each defendant to tract-level median income from ACS, based on the year their case was filed. For cases filed in 2004, we use pooled ACS data from 2004-2005; for cases in 2009-2013, we use pooled ACS data from 2009-2013; and for cases in 2014-2015, we use pooled ACS data from 2014-2015. We are able to geocode and assign a tract income for approximately 75% of cases.

B.2 Breath Test Incidents

We also obtained data on all breath test incidents in Texas from 2004 and 2009 through the middle of 2015. These records do not capture portable breathalyzer tests that occur during traffic stops. In fact, such portable tests are not admissible as evidence in Texas. Instead, these records comprise incidents in which the driver was detained, brought to a testing facility, and submitted to a breath test using a standardized and calibrated testing instrument named an "Intoxilyzer." The testing instrument outputs two readings for the individual's breath alcohol content (BrAC). We observe both readings, and although we focus on the lower of the two, the correlation between them is 0.94 and the readings are exactly the same in 95% of incidents.

In addition to exact BrAC readings, the data we received via public records request also include the driver's full name, date of the test, driver's age at the time of the test, and the county in which the test occurs. Given the date of the test and the driver's age at the time of the test, we can construct a one-year range of plausible birth dates for each driver. Since we aim to link these incidents with the detailed court records in Harris County, we restrict attention to the $48,316$ breath tests that occur in Harris County over this period.

B.3 Linking Breath Tests to Court Records

Initially, we conduct a cross-product merge of the breath test records from Harris County to the court records on the basis of first and last name. Matching on name alone will yield many false positives, so we implement a series of restrictions to ensure our matched records correctly link drivers to defendants. First, we eliminate any "match" for which the defendant's exact birth date does not fall within the plausible range of birth dates for the driver. Second, we drop any match for which the case filing date is before the test date or more than two days after the test date. Next, we use additional name information such as middle initial and suffix to further filter out matches that are likely false positives. For example, we remove matches where the middle initial of the defendant is not the same as the middle initial of the driver. The resulting file contains a match for 79% of the $48,316$ breath test incidents in Harris County, and we view these as high-quality matches since they pass all of our restrictions outlined above.

Next, we attempt to link the unmatched breath test incidents by relaxing the exact name matching implemented above. We conduct a cross-product merge of the unmatched breath test incidents to the court records on the basis of the first three letters of the individual's first name and the first three letters of the individual's last name. We again require that the defendant's exact birth date fall in the range of plausible birth dates for the driver, and again remove any matches for which the case file date occurs before the test date or more than two days after. Finally, we calculate the Jaro-Winkler string distance between the full name of the defendant and the full name of the driver, and remove matches that are very dissimilar. We find an additional $1,648$ high-quality matches from this process, increasing our overall match rate to approximately 82%. Figure A.1 displays the match rate (i.e., "share charged") by BrAC.

B.4 Constructing our Analysis Sample

The matched file contains any DWI breath test incident from Harris County that results in a charge in Harris County District Courts. For the purposes of this paper, we want to narrow the incidents under consideration to make our comparisons across defendant characteristics as exact as possible. The primary way we do this is by controlling for the individual's BrAC in each case. However, there are other legally-relevant factors that we control for by restricting our sample.

First, we limit our sample to individuals without a prior criminal history. Sanctions for DWI are explicitly a function of prior DWIs, and judges or prosecutors may take into account one's broader criminal history. We focus on first-time offenders by removing all cases where we identify a prior charge in Harris County going back at least fourteen years. We also drop cases in which the offense string indicates the charged offense is a "2ND" or "3RD" DWI.[1]

Second, we only consider cases in which the breath test incident results in a single DWI

[1] We explore robustness to more stringent criminal history controls in Table A.3 and Figure A.7.

charge. This excludes, for example, individuals who are charged with DWI and "reckless driving." We also eliminate cases in which the charged offense is "DWI with child passenger" and cases in which the defendant is below the age of 21. Finally, we remove individuals who are not U.S. citizens per the court records, to avoid conflating disparities by race/ethnicity with immigration status. These four restrictions remove cases with various aggravating circumstances.

Last, we remove all cases in which the individual's BrAC is below the legal limit or is missing. In these cases, we do not have an objective measure of guilt. In the small number of cases in which BrAC is missing, we do not know the individual's level of intoxication. When BrAC is below the legal limit, guilt may be determined by other toxicology results, by more subjective witness testimony, or other unobserved factors. It is only in cases above the legal limit where we can say definitively that the individual meets the legal standard for guilt. The resulting file, after these restrictions, contains $12,888$ first-time DWI cases above the legal limit and with no aggravating factors.